

Credit Commitments by Nonbanks*

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Abstract

Despite the secular shift to nonbank lending, banks retain a distinctive advantage in providing credit lines (Kashyap, Rajan, and Stein, 2002). We document a new pattern in nonbank lending: business development companies (BDCs) extend substantial credit commitments to borrowers, with commitment-to-asset ratios comparable to those of banks. These off-balance-sheet commitments are concentrated, exposing BDCs to undiversified borrower liquidity shocks—a risk they manage with credit lines provided by banks. We develop a model of layered liquidity insurance along the credit chain: the BDC pools borrower firms to partially diversify idiosyncratic shocks and uses a bank credit line to backstop residual liquidity risk. In equilibrium, the nonbank optimally provides only partial liquidity insurance, which generates a novel externality whereby firms sharing the constrained liquidity pool inefficiently underinvest in liquidity management.

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1 Introduction

The secular decline in banks’ balance-sheet lending has blurred the boundaries between banks and nonbanks in financial intermediation (Buchak, Matvos, Piskorski, and Seru, 2024). For example, private credit has grown rapidly in middle-market corporate lending, the nonbank share of residential mortgage originations has increased substantially, and fintech lenders have expanded in small-business lending.¹ Yet banks retain a distinctive advantage in providing credit lines, owing to their access to deposit funding (Kashyap, Rajan, and Stein, 2002). Credit lines insure borrowers against liquidity shocks and thereby improve real efficiency (Holmström and Tirole, 1998), but providing such insurance is challenging for nonbanks, which lack stable deposit funding.

In this paper, we document a new pattern in the provision of liquidity insurance. Using novel hand-collected data, we show that business development companies (BDCs), a large segment of nonbank lenders, issue substantial credit commitments to their portfolio borrowers in the form of revolving credit lines and delayed-draw term loans. We find that these off-balance-sheet commitments have grown rapidly in recent years and become economically significant relative to BDCs’ total assets, with commitment-to-asset ratios reaching levels comparable to those of banks by the end of 2025. This pattern suggests that credit commitments provided by nonbanks have become an increasingly important source of liquidity insurance to corporate borrowers.

A natural question is how BDCs provide these credit commitments and manage the associated liquidity risk. As Holmström and Tirole (1998) show, when liquidity shocks are purely idiosyncratic and fully diversifiable, any intermediary with commitment power can provide liquidity insurance by pooling borrowers and raising funding from spot markets. Do

¹See Block, Jang, Kaplan, and Schulze (2024); Buchak, Matvos, Piskorski, and Seru (2018); He, Huang, and Zhou (2023); Huang (2021).

BDCs simply diversify idiosyncratic shocks, or do they ultimately rely on a stable source of funding from other intermediaries? To answer this question, we combine detailed facility-level data on BDCs' commitment books with their loan portfolio holdings, balance sheets, cash flow statements, and importantly, off-balance-sheet credit lines that BDCs receive from third-party lenders, which are mostly banks.

We document two empirical facts. First, BDCs' credit commitments expose them to large and volatile liquidity shocks. We measure the liquidity shock a BDC experiences in a quarter as the change in unfunded commitments (i.e., undrawn credit extended by the BDC) across the borrowers in its commitment book, and its volatility as the time-series standard deviation of this shock. A typical BDC's volatility is about two and a half times as large as that of an equal-weighted benchmark formed as the aggregate of all BDCs' unfunded commitments, and even the least volatile quartile of BDCs is nearly twice as volatile as the benchmark. These shocks are also sizable: a one-standard-deviation quarterly shock could deplete the BDC's cash holdings in about one-fifth of BDC-quarters, and the magnitude is even more pronounced for BDCs with greater commitment books.

Two economic forces give rise to these large and volatile shocks: book concentration and shock commonality. BDC commitment books are substantially more concentrated by borrower industry, geography, and sponsorship, relative to their on-balance-sheet loan portfolios. This concentration reflects lender information in specific borrower segments, as the commitments also exhibit stronger segment specialization than loan holdings. Within each segment, BDCs further selectively extend commitments to borrowers with better credit quality, sponsor backing, dual debt-and-equity exposure, and recent acquisition activity, suggesting borrower-level risk considerations. Moreover, borrower liquidity shocks also carry a common component, with the 2020Q1 COVID episode as the most visible example. A concentrated book with correlated borrower liquidity shocks leaves the BDC with undiversified exposures.

Second, BDCs manage this liquidity risk through bank credit lines they receive as borrowers. In the cross section, BDCs with larger commitments to borrowers reserve larger undrawn bank credit lines. These undrawn bank credit lines, not cash, provide a liquidity backstop. When cash needs arise, BDCs not only draw cash from their existing bank credit lines but also expand their committed capacity, by either amending credit limits upward or entering into new facilities. This backstop is also resilient to stress: the credit limits to BDCs did not contract when the banks were under pressure, such as during the 2023 regional bank crisis. However, this chain is a potential source of fragility amplification that regulators have flagged—the [FSOC 2024 Annual Report](#) (Box E) warns of a “dash for liquidity” in which portfolio firms draw on their direct lenders, who in turn draw on their bank facilities—to our knowledge, we offer the first study to document it in the data.

With BDCs as a laboratory, our evidence raises deeper questions about the growing provision of liquidity insurance by nonbanks. Given nonbanks’ exposure to undiversified borrower liquidity shocks and their reliance on bank credit lines, does the liquidity insurance they provide differ fundamentally from that of traditional banks? And if so, how does this difference shape the behavior of the borrowers they insure?

To answer these questions, we develop a model of layered credit-line provision that builds on [Holmström and Tirole \(1998\)](#). Firms face liquidity shocks with both idiosyncratic and aggregate components and endogenously choose their exposure through costly risk-management effort. Because spot markets react to ex post liquidity shocks and self-insurance through cash is costly, firms seek credit lines for liquidity management. Motivated by our evidence, a BDC extends credit to a finite number of firms and provides a first layer of insurance by partially pooling their idiosyncratic shocks. Because this leaves undiversified liquidity risk, the BDC relies on a committed credit line from a bank to fund borrower drawdowns.

Banks provide a second layer of insurance by committing liquidity to BDCs across states

of portfolio drawdowns. Regulatory capital rules disincentivize banks from lending to middle market firms directly, so they instead lend to them indirectly through the credit chain. Holding large and diversified portfolios of exposures, banks fully diversify idiosyncratic risk across intermediaries, so that only aggregate liquidity risk remains at the bank level. We introduce “sleepy” depositors who supply funding ex ante and do not fully price aggregate risk ex post, allowing the banking sector to absorb aggregate liquidity shocks.

The model delivers two implications unique to nonbank liquidity provision. First, because the BDC is subject to undiversified borrower liquidity risk and has no public backstop against widespread drawdowns, it optimally rations its liquidity provision to limit potential bankruptcy costs. In the model, rationing does not mean excluding borrower firms, which would forgo the gains from pooling; rather, the BDC takes a smaller bank credit line—effectively choosing limited commitment power ex post over the credit lines extended to borrower firms. Our data bear out this prediction: a one-dollar increase in unfunded commitments to borrower firms is associated with only about \$0.53 of undrawn bank credit slack, and the slack available to cover drawdowns is smaller still because BDCs also tap these lines for other liquidity needs.

Second, this constrained insurance generates an *externality* among the firms sharing the pool—in essence, a tragedy of the commons. Because drawdowns in widespread-shock states are rationed by a limited bank credit line, a firm that exerts less effort in liquidity management does not internalize that its own increased liquidity risk and drawdown on the credit line would crowd out coverage for the others. Therefore, in the decentralized equilibrium, firms inefficiently exert less effort in liquidity management relative to the socially optimal level. This externality is specific to nonbanks’ constrained liquidity provision: it arises only when the pool is rationed. Banks, by contrast, hold diversified books, access deposit funding, and expect intervention in a crisis, so their committed capacity rarely binds

and the externality does not arise. It is worth comparing this to [Bräuning and Ivashina \(2026\)](#), who identify a related liquidity-pool problem for banks: they study an *ex-post* run on committed credit lines in a crisis, when firms draw on their facilities at once and strain their lenders. The inefficiency we identify is instead *ex ante* on the inefficient allocation that firms seeking nonbank liquidity insurance exert less effort in liquidity management than the socially optimal level.

Related Literature

Our paper relates to several strands of literature, most closely the literature on *credit lines and bank liquidity provision*. Extending the diversification logic of [Diamond \(1984\)](#) to liquidity provision, [Holmström and Tirole \(1998\)](#) show that credit lines deliver the second-best allocation under idiosyncratic liquidity shocks.² Banks, with stable funding supplied by “sleepy” depositors ([Hanson, Shleifer, Stein, and Vishny, 2015](#)) and, in crisis, government support ([Acharya and Mora, 2015](#)), are argued to have a structural advantage in providing credit lines. In fact, liquidity provision is widely identified as a defining advantage of banks over nonbank intermediaries, and rationalizes the very existence of bank balance-sheet lending which combines credit extension with payment services ([Kashyap, Rajan, and Stein, 2002](#); [Gatev and Strahan, 2006, 2009](#); [Pennacchi, 2006](#)). Connecting this bank-centered view to the post-GFC migration of credit from banks to nonbanks ([Buchak, Matvos, Piskorski, and Seru, 2018, 2024](#); [Irani, Iyer, Meisenzahl, and Peydró, 2021](#)), we study the growing extension of credit commitments by nonbanks and its implications for liquidity provision.³

²For other theories of credit lines, see [Acharya, Almeida, Ippolito, and Perez-Orive \(2014\)](#) on monitored liquidity insurance and [Donaldson, Koont, Piacentino, and Vanasco \(2024\)](#) on the role of contingent commitment in mitigating debt dilution.

³Consistent with this view, nonbank lenders typically extended corporate credit in the form of term loans instead of credit commitments ([Bord and Santos, 2012](#); [Berlin, Nini, and Yu, 2020](#); [Haque, Mayer, and Stefanescu, 2024](#)), and existing studies on nonbank lending have largely focused on the origination and secondary market trading of term loans (e.g., [Nadauld and Weisbach, 2012](#); [Lim, Minton, and Weisbach, 2014](#); [Bruche, Malherbe, and Meisenzahl, 2020](#); [Bruche, Meisenzahl, and Xu, 2026](#); [Ivashina and Sun, 2011](#); [Elkamhi and Nozawa, 2022](#); [Xu, 2026](#)).

We also speak to the *empirical literature on credit lines* (e.g., [Sufi, 2009](#), [Greenwald, Krainer, and Paul, 2025](#); see [Demiroglu and James, 2011](#) for a review). Most relevant to our paper is whether credit lines cover idiosyncratic or aggregate liquidity shocks, because diversifying idiosyncratic shocks only requires an intermediary with commitment, not necessarily a bank. Along this line, [Acharya, Almeida, and Campello \(2013\)](#) point to a prioritization of idiosyncratic shocks: firms with greater aggregate-risk exposure obtain credit lines on worse terms and substitute toward cash. Aggregate shocks are nonetheless covered to some degree as evidenced by [Acharya, Engle, Jager, and Steffen \(2024\)](#): when the aggregate COVID-19 shock hit, banks that had committed credit lines underperformed in the stock market.

Our focus on BDCs’ reliance on bank credit lines as a liquidity backstop connects to the literature on *intermediation chains*. [Acharya, Gopal, Jager, and Steffen \(2024\)](#) document that bank lending to nonbank financial intermediaries has grown rapidly, mainly in the form of committed credit lines (see also [Acharya, Cetorelli, and Tuckman, 2026](#)). As the upstream layer of the chain, bank credit has been studied as a key funding source for mortgage originators ([Jiang, 2023](#)), BDCs ([Chernenko, Ialenti, and Scharfstein, 2025](#); [Haque, Jang, and Wang, 2025](#)), and nonbanks more broadly ([Xu, 2025](#)). The literature offers several explanations for why credit chains arise, including regulatory arbitrage ([Chernenko, Ialenti, and Scharfstein, 2025](#)), asymmetric information ([Glode and Opp, 2016](#)), and insulation against rollover risk ([He and Li, 2024](#)). While this literature has studied banks’ liquidity provision to nonbanks, our contribution is a *two-layered* liquidity-insurance chain in which the nonbank’s own liquidity need is endogenous to the commitments it has extended to borrower firms.

The implication that nonbanks’ constrained liquidity pool induces lax liquidity management is, in essence, a tragedy of the commons ([Hardin, 1968](#)). It connects to [Bräuning and Ivashina \(2026\)](#) and [Ivashina and Scharfstein \(2010\)](#): theirs is an *ex-post* run on committed capacity, whereas ours is an *ex-ante* distortion in firms’ liquidity management.

Finally, we use BDCs as a laboratory to study nonbanks and hence connect to the growing *literature on private credit and BDCs*. This literature has studied private credit lenders’ balance sheets (Matvos, Piskorski, and Seru, 2026), lending technology (Jang, Kim, Sufi, and Chen, 2025), loan contractual terms (Chernenko, Erel, and Prilmeier, 2022; Block, Jang, Kaplan, and Schulze, 2024; Jang, 2024), as well as lending dynamics and market discipline (Davydiuk, Marchuk, and Rosen, 2024a,b). Notably, Haque, Mayer, and Stefanescu (2024) document a within-borrower complementarity between term loans provided by nonbanks and credit lines provided by banks. Our paper contributes to this literature by examining private credit lenders’ liquidity provision and off-balance-sheet commitments.

2 Data

2.1 Institutional Background

Business development companies (BDCs) are closed-end investment vehicles regulated under the Investment Company Act of 1940. To be eligible, a BDC must hold at least 70% of its assets in qualifying assets, predominantly small and middle-market private U.S. operating companies. BDCs invest across the corporate capital structure but concentrate in directly originated senior secured loans. Most BDCs elect to be taxed as regulated investment companies (RICs), which avoid corporate-level tax by distributing at least 90% of their investment company taxable income to shareholders each year. Our study focuses on public BDCs, which are subject to public disclosure requirements and file periodic reports with the SEC.⁴ These BDCs consist of exchange-listed BDCs, whose equity shares trade on exchanges, and non-traded BDCs, which offer only periodic share repurchases, commonly capped near 5% of net asset value per quarter.

⁴We exclude private BDCs, which rely on private market investors and are exempt from these disclosures.

BDCs are now a large segment of the private-credit market, which is one of the fastest-growing parts of the financial system, now with over \$1 trillion in the United States.⁵ The universe of BDCs we identify from SEC EDGAR comprises 245 unique BDCs that are ever public between 2005Q1 and 2025Q4, with combined total assets of roughly \$550 billion as of 2025Q4. These public BDCs offer a transparent window into the opaque private-credit market. Figure [IA.4](#) in the Internet Appendix shows the total number and assets of these listed and non-listed BDCs.

2.2 Data Sources

We assemble four datasets covering business development companies (BDCs): (i) a BDC-quarter panel of balance-sheet, cash-flow, and off-balance-sheet commitment items; (ii) the unfunded-commitment book at the BDC-quarter-borrower-facility level, covering the credit commitments BDCs extend to portfolio companies; (iii) a quarterly panel of bank credit lines that BDCs obtain as borrowers; and (iv) loan portfolio holdings at the BDC-quarter-borrower-facility level. Datasets (i)–(iii), which cover BDCs’ commitments and their bank funding, are hand-collected from SEC EDGAR filings; we supplement them with data from PitchBook, FactSet, Compustat, S&P Capital IQ (CIQ), and the FFIEC Call Report. Internet Appendix [IA.6](#) provides details on the collection and processing of these datasets.

BDC-quarter panel. We identify the universe of public BDCs through Form N-54A, N-54B, and N-54C filings on EDGAR, which respectively elect, amend, and withdraw BDC status under the Investment Company Act of 1940. The sample consists of 6,747 BDC-quarters across 245 unique BDCs over 2005Q1–2025Q4 in which the BDC has an active N-54A election. The main BDC-quarter variables are (a) the BDC’s total unfunded commit-

⁵See “[Bank Lending to Private Credit: Size, Characteristics, and Financial Stability Implications](#),” FEDS Notes, Federal Reserve Board, May 2025.

ments (the credit committed but undrawn) to portfolio companies; (b) balance-sheet items including total assets, total debt, total equity, and cash and equivalents; and (c) statement-of-cash-flow items including operating, investing, and financing cash flow subtotals together with gross investment purchases, gross investment sales and repayments, distributions paid, and equity issuance.

Unfunded credit in commitment books. For the subsample of BDC-quarters in which the filing’s commitments-and-contingencies note discloses a borrower-level breakdown of the unfunded loan commitments, we hand-collect each portfolio company’s unfunded amount, the type of credit commitment (revolving credit facility, delayed-draw term loan, or other), and other reported terms. We use DDTL to refer to “delayed-draw term loan” henceforth. The resulting dataset covers 2,989 BDC-quarters and 138,456 facility-level observations. For analyses that use the change in unfunded commitment as a proxy for drawdown activity, we restrict the sample to borrower-facility-quarters with lagged unfunded above \$1 million to avoid small-denominator noise. To identify borrowers in these commitment books and merge their firm characteristics, we link each borrower to a PitchBook company identifier (see Internet Appendix [IA.6.6](#)).

Quarterly panel of bank credit lines. BDCs disclose each revolving facility they obtain as borrowers in a nonstandardized textual borrowings note; because the lead lenders are predominantly banks, we refer to these facilities as bank credit lines. We assign a credit-line identifier (CLID) to each facility and hand-collect, at each filing date, the credit limit, drawn balance, undrawn capacity, lender identities (administrative agent and other syndicate members), interest-rate terms (benchmark and spread), maturity, and any contractually specified accordion provision. We also build a complementary event panel recording each facility’s inception, amendments, accordion exercises, terminations, and replacements. The CLID-quarter panel covers 9,055 CLID-quarters across 756 unique CLIDs and 198 BDCs

over 2005Q1–2025Q4. Internet Appendix [IA.8](#) summarizes the lender composition, the event panel, and the prevalence of accordion features.

Portfolio holdings. We obtain data on BDC portfolio holdings (term loans, revolvers, DDTLs, bonds, equity, structured products, and other debt) at the BDC-quarter-borrower-facility level from PitchBook, linked to BDCs via SEC Central Index Keys (CIKs). Holdings data are available for 5,769 BDC-quarters in our sample. We also draw on portfolio holdings from FactSet to facilitate linking borrowers between the commitment book and the portfolio holdings datasets.

Supplementary data. We use the FFIEC’s Consolidated Reports of Condition and Income (Call Reports) to construct the bank-quarter variables. We use S&P Capital IQ Capital Structure data merged with Compustat quarterly to construct aggregate public-firm draw-down factors, with details in Internet Appendix [IA.7.2](#). We also cross-validate the hand-extracted BDC balance-sheet items against Compustat and CIQ where coverage permits.

2.3 Summary Statistics

Table [1](#) summarizes the BDC-quarter panel. Panel A presents all 6,747 BDC-quarters. The average BDC holds \$1.5 billion in total assets, with a median of \$484 million and the 95th percentile exceeding \$5.8 billion. BDCs use relatively low financial leverage: average equity is \$811 million, against average total debt of \$663 million. As lenders, the average BDC extends \$148 million of total unfunded commitments to borrowers, with a large right tail reaching \$531 million at the 95th percentile. As borrowers, BDCs receive on average 1.4 bank credit lines with an average total credit limit of \$566 million, of which on average \$261 million remains undrawn.

Panel B restricts to the 2,977 BDC-quarters that disclose a borrower-level commitment book. The average BDC reports 44 outstanding unfunded commitments to portfolio compa-

nies, with an average unfunded amount per commitment of \$4.9 million (median \$3.4 million). These magnitudes are consistent with BDCs lending to middle-market borrowers below the size of typical syndicated loans. Panel C reports the fair-value composition of BDC portfolio holdings. On average, senior secured loans (term loans, unitranche, and the drawn portion of revolvers and DDTLs) account for 74% of holdings. Equity stakes in operating portfolio companies account for 12%, joint-venture lending vehicles (e.g., senior loan funds and specialty-lending platforms, identified by issuer-name patterns) for about 1%, and other debt (subordinated, mezzanine, and unsecured) for the remaining 14%.

Table 2 summarizes the commitment book at the BDC-facility-quarter level. The sample contains 66,325 facility-quarters with at least \$1 million of lagged unfunded amount. Panel A presents all commitment types. The average unfunded amount is \$7.0 million, with a median of \$3.2 million and a right tail reaching \$23.8 million at the 95th percentile. The average dollar change in unfunded amount is $-\$0.4$ million per quarter, and the average percentage change is -2.4% , both consistent with net drawdown on average. Borrowers actively draw down—that is, the unfunded amount decreases—in 34.5% of facility-quarters. The average time to maturity is 3.3 years. Panels B and C present revolvers and DDTLs separately. Revolvers are smaller than DDTLs on average (\$5.0 million vs \$8.4 million) but have a longer maturity (4.2 years vs 2.3 years). DDTLs are drawn more frequently (41.5% of commitment-quarters) than revolvers (26.0%), consistent with DDTLs serving as acquisition-financing facilities with scheduled tranche draws.

3 Empirical Patterns

We document three empirical patterns in this section. First, BDC credit commitments are an economically important and growing source of corporate liquidity insurance. Second, BDCs’ commitment books expose them to volatile borrower liquidity shocks. Finally, BDCs

manage this liquidity risk using bank credit lines as a liquidity backstop. Together, these patterns motivate the credit chain that our model formalizes in Section 4, as well as its key premise that BDCs’ liquidity risk from borrowers is undiversified.

3.1 Economic Significance of BDCs’ Unfunded Commitments

BDC unfunded commitments to portfolio companies are an economically important source of liquidity insurance. Figure 1, Panel A compares the total unfunded commitments BDCs extend to portfolio companies with those banks extend to corporate borrowers (commercial and industrial firms and financial institutions), both scaled by the sector’s total assets. The commitment-to-asset ratio of BDCs has been rising over the past decade, and by 2025Q4 the two ratios are comparable at about 15%. Panel B shows the cross-sectional distribution of this ratio as of 2025Q4. Whereas most banks have modest ratios, typically below 10%, about 60% of BDCs have a ratio above 10%, suggesting that liquidity insurance provision is now widespread within the BDC sector. We are aware that this ratio is affected by bank versus BDC asset composition; nonetheless, it is a valid measure to show that BDCs provide a significant amount of corporate liquidity insurance.

Figure 2 plots, in aggregate, the liquidity BDCs provide to portfolio companies and the liquidity BDCs can themselves access. First, the total amount of BDC unfunded commitments grew rapidly, from near zero in 2005 to roughly \$80 billion by 2025. Second, this growth is broad-based: the number of BDCs with unfunded commitments has grown alongside the aggregate. Third, the liquidity BDCs provide is comparable in scale to the liquidity available to them: aggregate undrawn bank credit line capacity grows alongside their unfunded commitments to portfolio companies, and BDC cash holdings provide a smaller additional buffer. This parity between liquidity provided and liquidity available, together with their growing relevance, motivates us to write a credit chain model of two-layered credit

line insurance, which we formalize in Section 4.

3.2 BDC Exposure to Borrower Liquidity Shocks

In this subsection, we quantify liquidity shocks from BDCs' commitment books and identify two economic forces that drive the BDC-level exposure to those shocks. Throughout, let $u_{i,\ell,t}$ denote BDC ℓ 's unfunded commitment to borrower i in quarter t . Given the nature of the data (see Internet Appendix Section [IA.7.1](#) for details), we proxy for cashflows related to drawdown activity using changes in unfunded commitments of continuing positions that are observed in two consecutive quarters. We use two alternative measures of borrower liquidity shocks: a continuous percentage change $\Delta u_{i,\ell,t}/u_{i,\ell,t-1}$, and a binary indicator $\mathbf{1}\{\Delta u_{i,\ell,t} < 0\}$ for whether the borrower draws cash from the commitment.

3.2.1 Quantifying Liquidity Shocks from Commitment Books

BDC commitment books experience large quarterly liquidity shocks, well beyond what a fully diversified book of commitments would experience. To quantify these BDC-level shocks, we restrict the sample of BDC-borrower-quarter observations to BDC-quarters with at least five borrowers and to BDCs with at least 12 qualifying quarters. For each BDC in this sample, we compute the time-series standard deviation of its quarterly liquidity shock. We then compare this BDC-level volatility to a fully diversified benchmark, defined as the volatility of an equal-weighted book of all BDCs' commitments over the same time window.

Table [3](#) reports the results. In Panel A, the average BDC's volatility is 14.5% per quarter on the continuous measure and 14.2% on the binary measure, approximately 3 times as large compared with 4.8% and 5.7% for the diversified benchmark. In the cross section, the BDC-level volatility is consistently above the benchmark. This volatility gap indicates that borrower liquidity shocks do not fully diversify at the BDC level; instead, BDCs are exposed

to volatile liquidity shocks driven by their borrowers’ drawdown activity.

Panel B translates this volatility into dollar magnitudes by multiplying each BDC’s time-series standard deviation of liquidity shock by its lagged total unfunded commitments.⁶ The estimated dollar volatility is large: the shock equals 29.1% of the BDC’s lagged cash holdings for the median BDC-quarter, and it exceeds the BDC’s cash holdings in 20.6% of BDC-quarters. The magnitude is substantially larger for BDCs with greater commitment books: in the top tercile of BDC-quarters, the estimated volatility equals 60.5% of cash at the median and exceeds cash in 34.4% of BDC-quarters.

3.2.2 Concentration of Commitment Books

Book concentration is the primary reason of the large BDC-level volatility. We construct a sample of 2,619 BDC-quarters with data on both commitment books and portfolio holdings and separately compute concentration measures for these off- and on-balance-sheet exposures.⁷ At the borrower’s industry, headquarters-state, and PE sponsor levels, we use the Herfindahl–Hirschman Index (HHI) and top-three shares as concentration measures. Table 4 shows that BDC commitment books are substantially more concentrated than the same BDC’s portfolio holdings. Specifically, the median HHI is 0.23 versus 0.13 at the industry level, 0.18 versus 0.08 at the headquarters-state level, and 0.13 versus 0.04 at the sponsor level. The same pattern holds when concentration is measured by top-three shares. The differences in average concentration measures are all statistically significant.

Why do BDCs hold such concentrated commitment books? One reason is *specialization*: BDCs may deliberately over-weight industries, geographies, or sponsor relationships for which they have better information. Following [Blickle, Parlato, and Saunders \(2026\)](#),

⁶This calculation estimates the size of the typical cashflow in the commitment book. It does not measure realized total BDC-quarter borrower drawdowns, which are not observable in the data.

⁷This sample includes all BDC-quarters with data on both commitment books and portfolio holdings for which we have information on the borrower’s industry, headquarters-state, and PE sponsor from Pitchbook.

we compute each BDC-quarter’s weight in a borrower industry, state, or sponsor bucket in excess of the cross-BDC aggregate, separately for commitment books and portfolio holdings. Table 5 shows that at the industry level, the median BDC’s largest excess share is 25 pp for commitment books versus 12 pp for portfolio holdings, and the cumulative excess share across the top three industries is 45 pp versus 25 pp. The same pattern holds for borrower geography (21 pp versus 8 pp for Top-1 and 41 pp versus 17 pp for Top-3) and sponsor relationships (22 pp versus 9 pp for Top-1 and 48 pp versus 19 pp for Top-3), with all differences significant at the 1% level. This result suggests that BDCs extend commitments to borrowers for which they have superior screening or monitoring capacity.

Beyond specialization, BDCs may also selectively extend commitments to borrowers within a given industry and geography. To investigate this selection, we regress an indicator for whether a BDC extends an unfunded commitment to an existing borrower on its firm characteristics and relationship with the BDC. To account for industry and geographical heterogeneities and time-varying BDC-specific effects, our regressions include industry-by-quarter, HQ-state-by-quarter, and BDC-by-quarter fixed effects.

Table 6 reports the estimation results. A strong predictor of the extension of a commitment is the borrower’s credit quality, implying that the BDC selectively extends commitments to borrowers with lower default risk.⁸ Sponsor backing adds 3.6 pp to the probability, consistent with evidence that sponsor engagement can substitute for lender monitoring (Haque, Mayer, and Wang, 2026). The Dual Held indicator adds 2.2 pp, and this relationship is concentrated in revolvers, consistent with evidence that BDC dual holders are compensated for monitoring and governance services (Davydiuk, Erel, Jiang, and Marchuk, 2025). Recent M&A activity adds another 4.7 pp to the probability, and its association is stronger on DDTLs, which are typically used for acquisition financing. Moreover, smaller

⁸We measure credit quality at the borrower-quarter level using the lagged cost-weighted average fair-value-to-cost ratio across all BDCs’ term-loan holdings of the borrower.

and younger borrowers are more likely to receive commitments, and the BDC’s larger existing exposure to the borrower also positively predicts commitment extension. Overall, these findings suggest that BDCs selectively extend commitments based on both borrower needs and risk considerations, contributing to the concentration of BDC commitment books.

3.2.3 Commonality of Borrower Liquidity Shocks

A secondary reason of the large BDC-level volatility is commonality in borrower liquidity shocks, which cannot be diversified away even with a perfectly diversified commitment book. Figure 3 plots the quarterly averages of the two borrower liquidity shock measures between 2017Q1–2025Q4. A salient event is the 2020Q1 COVID drawdown, where the average percentage change in unfunded drops to roughly -35% and the fraction of borrowers actively drawing exceeds 60%. Across commitment types, revolvers show a stronger COVID-quarter co-movement relative to DDTLs (Figure IA.1 in the appendix). These episodes of synchronized drawdown suggest the existence of commonality across borrowers.

Table 7 reports univariate panel regressions of the borrower liquidity shock on time-series factors. Panel A uses quarterly average percentage change in credit line drawn balance from S&P Capital IQ to capture the drawdown activity of public firms. We measure this activity separately for four lagged Debt/EBITDA buckets representing different credit profiles, with details in Internet Appendix IA.7.2. The estimates suggest that BDC borrowers’ liquidity shocks comove with the drawdowns of public firms, especially public firms with higher leverage or negative earnings. Panel B regresses the shock on market-wide macro and credit-cycle factors: changes in the BBB and high-yield credit spreads, the S&P500 return, and changes in the VIX. All estimates suggest that borrowers on commitment books pass correlated liquidity shocks to BDCs when equity and credit markets are under stress.

These panel regressions illustrate which factors the liquidity shocks co-move with. To

quantify the undiversified component of these shocks, we present a variance decomposition in Table 8 via different sets of fixed effects. Time fixed effects alone absorb roughly 1% of variation, suggesting that aggregate time variation in the shocks is small. However, industry-by-time and HQ-state-by-time fixed effects absorb 4–11% and 6–14% respectively, indicating that the shocks are correlated across borrowers within the same industry and geography. Combined with the book concentration documented in Section 3.2.2, these comovements give rise to the sizable BDC-level exposure to volatile liquidity shocks. Indeed, BDC-by-time fixed effects absorb 8–12%, suggesting that a substantial share of the shocks fail to diversify away within the BDC’s commitment book. This undiversified liquidity risk is the empirical counterpart of the key assumption in our model (Section 4): because a BDC pools only a finite, concentrated book of borrowers with correlated shocks, substantial liquidity risk remains at the BDC level. The next subsection shows that bank credit lines serve as a backstop for this residual risk, which the model casts as the second layer of insurance in the credit chain. In the model, because shocks in tail states can still threaten the BDC’s own survival, we show that the BDC may optimally provide only partial liquidity insurance.

3.3 Bank Credit Lines as a Liquidity Backstop

When borrowers draw their unfunded commitments, the BDC has to deliver cash, so the volatility documented in Section 3.2 gives rise to liquidity risk for BDCs. This subsection examines how BDCs satisfy these drawdown requests and manage the liquidity risk with bank credit lines.

3.3.1 Relationship Between Liquidity Provided and Liquidity Available

Figure 2 has shown that BDCs’ aggregate unfunded commitments to borrowers and their aggregate undrawn credit lines from banks have been growing together over 2005–2025 at

a similar scale. These credit lines provide an important source of liquidity to BDCs. As Figure 4 shows, the aggregate undrawn fraction of the total bank credit lines has remained stable at 45–50% over time, which provides a liquidity backstop for managing the liquidity shocks from the commitment books and other cash needs. Indeed, BDCs themselves identify bank credit lines as the primary backstop for their unfunded commitments in regulatory filings.⁹ Figure 5 reports the share of BDC 10-K filings naming each liquidity source as backup for unfunded commitments. Bank credit facilities (82.0%) and cash (75.2%) are the most frequently named sources, far ahead of operating cash flow, marketable securities, and LP capital commitments. Moreover, because BDCs must distribute most of their income to maintain tax status, they hold only a small cash buffer.¹⁰ So bank credit lines, rather than cash, are likely the most important liquidity backstop in terms of economic magnitude.

Next, we quantify the cross-sectional relationship between credit commitments the BDC extends to their borrowers and receives from banks. Specifically, we regress a BDC’s bank credit line quantities on the BDC’s total unfunded commitment to borrowers in its commitment book, controlling for the BDC’s lagged total assets, cash and equivalents, and total debt, all scaled by lagged total assets. We also control for a dummy for the BDC’s public status. The regressions include different sets of fixed effects to account for time and BDC-level heterogeneity. The sample consists of BDC-quarters with a bank credit line, consistent with the model’s focus on BDCs that use bank credit lines as a liquidity backstop.

Table 9 reports the results. Columns 1–3 and 4–6 use the total bank credit limit and the undrawn amount as the dependent variables, respectively. Across all columns, the coefficient on total unfunded commitment is positive and statistically significant, indicating that BDCs with larger commitments to borrowers also have larger bank credit lines from banks. The

⁹We classify 1,243 BDC 10-K filings and restrict to the 605 filings that explicitly link a liquidity source to unfunded commitments. Internet Appendix Figure IA.2 reports disclosure status over the full sample.

¹⁰Internet Appendix Figure IA.3 shows that the ratio of BDCs’ undrawn bank credit and cash holdings is several times that of both nonfinancial and financial public firms.

coefficient is economically large: in the most stringent specifications (columns 3 and 6), a one-dollar increase in unfunded commitments is associated with about \$0.90 of additional bank credit line capacity, of which roughly \$0.53 is held as undrawn slack. The undrawn slack is well below dollar-for-dollar, and because BDCs also tap these lines for other liquidity needs such as new investment, the slack available to cover commitment drawdowns is even smaller. Bank credit lines are thus unlikely to fully cover widespread drawdowns—the BDC appears to provide only partial liquidity insurance, a pattern our model in Section 4 rationalizes.

3.3.2 BDC Liquidity Needs and Bank Credit Lines

This subsection examines how BDCs’ bank credit lines respond on the margin to BDC liquidity needs. Ideally, we would measure borrower total drawdowns at the BDC-quarter level. However, we cannot cleanly measure this BDC-level total amount because changes in the total unfunded commitment are confounded by new originations and terminations, as well as amendments that increase or decrease the total commitment size.¹¹ Moreover, in the cash flow statement, borrower drawdowns on unfunded commitments are reported together with other investment activities under a single cash outflow item. We therefore use the BDC’s gross investment cash outflow as a measure to capture liquidity needs the BDC faces in a given quarter. Internet Appendix Table IA.4 illustrates the cash-flow items used in the analysis with a concrete example: Ares Capital around the 2020Q1 COVID stress.

We estimate the association between the BDC’s liquidity needs and its bank credit line activities. In column 1 of Table 10, we regress the amount of the BDC’s bank credit line drawdown on its quarterly gross investment cash outflow, both scaled by lagged total assets. We control for the BDC’s undrawn credit limit, total debt, cash and equivalents, and log total

¹¹The facility-level measure in Section 3.2 is not subject to this confounding: the sample is restricted to continuing facilities with at least \$1M lagged unfunded (Internet Appendix IA.7.1), which isolates drawdown activity from origination and termination effects.

assets. The regression includes BDC fixed effects to account for time-invariant BDC-level heterogeneity and year-quarter fixed effects to absorb common time trends in bank liquidity provision. The coefficient on the gross investment cash outflow is positive and statistically significant, suggesting that a one-dollar increase in total liquidity needs is associated with 10 cents of drawdown on the BDC’s existing bank credit lines.

In columns 2-4, we consider three other margins on which BDCs and banks adjust the credit lines. Column 2 examines amendments to existing facilities, where the dependent variable is the change in credit limit. Column 3 examines new facilities, where the dependent variable is the credit limit of newly originated facilities. The coefficient on the gross investment cash outflow is positive and statistically significant for both amendments and new originations. For each dollar of cash outflow, the BDC obtains 4 additional cents of existing facility limits and originates 6 cents of credit limits from new facilities. Column 4 examines the change in credit limit due to terminations of existing facilities, where the coefficient is small in magnitude and statistically insignificant. In total, the BDC expands its bank credit line capacity by 20 cents per dollar of incremental investment cash outflow.¹²

Taken together, Tables 9 and 10 provide suggestive evidence that BDCs use bank credit lines as a liquidity backstop. This backstop motivates the second layer of liquidity insurance provided by banks in our model (Section 4). In levels, BDCs maintain undrawn bank credit lines commensurate with their unfunded commitments to borrowers in their commitment books. In flows, BDCs expand their bank credit line capacity along with investment cash outflow. Consistent with this backstop role, BDCs’ bank credit lines are highly resilient to stress: Internet Appendix Table IA.3 shows no contraction in committed bank credit limits for BDCs more exposed to regional or low-capital banks around the 2023 regional bank crisis.

¹²A caveat is that we do not observe clean exogenous shocks to the BDC’s liquidity needs. Hence, the estimates should be interpreted as associations driven by the BDC’s investment and financing choices. Table IA.2 in the Internet Appendix shows that the gross investment cash outflow is also positively associated with its cash inflows from other funding sources such as investment payoffs and equity issuance.

4 A Model of Layered Credit Line Provisions

We develop a model of an intermediation chain in which a bank extends a credit line to a BDC, which in turn provides credit commitments to borrower firms. Specific to the nature of credit lines, our model highlights two layers of insurance: first, the BDC pools borrower firms and partially diversifies idiosyncratic liquidity shocks; second, the bank insures the BDC against residual liquidity risk—aggregate shocks are absorbed by sleepy depositors, while the remaining idiosyncratic component is diversified across the bank’s portfolio.

4.1 Model Setup

There are three dates, $t = 0$, $t = 1$ and $t = 2$, and there is no discounting. All agents are risk-neutral.

Firms. Firms are ex ante identical. At $t = 0$, each firm needs to finance 1 dollar to launch a long-term project. At the interim date $t = 1$, the firm is subject to a liquidity shock with probability p . If the shock occurs, the firm must raise an additional $\rho > 0$ to continue operating; absent this liquidity injection, the project is liquidated and yields zero. The liquidity shock is verifiable, so a firm cannot pretend to be shocked and divert cash when interim credit is available. The distribution of liquidity shocks across firms will be discussed later. If continued, the project produces $y > 1$ at $t = 2$.

A portion Δ of the firm’s output y is non-pledgeable, which constrains its ability to raise sufficient funds from the spot market at $t = 1$ and makes liquidity management relevant. One microfoundation is moral hazard, under which Δ is the non-pledgeable rent to the

entrepreneur to incentivize effort.¹³ We assume that

$$y - \Delta < \rho, \tag{1}$$

so that a firm experiencing a liquidity shock cannot raise sufficient funds from the spot market at $t = 1$ to cover the required liquidity injection ρ , even via fully senior claims that entirely dilute existing $t = 0$ creditors.

Self-insurance through cash savings is inefficient because holding cash has an opportunity cost: the firm pays the required return on every dollar raised, and the inefficiency arises when no liquidity shock hits and the precautionary buffer sits idle. Specifically, to self-insure, the firm must raise $1 + \rho$ at $t = 0$. Let $R > 1$ denote the gross return on outside investment. We assume that

$$y - \Delta + (1 - p)\rho \leq (1 + \rho)R, \tag{2}$$

so that the firm cannot raise sufficient funds for self-insurance. This assumption allows us to focus on the provision of credit lines.

Firms may seek financing either directly from the bank or from the BDC. In this paper, we focus on the intermediation chain allocation in which firms borrow from BDCs, and BDCs in turn borrow from banks. As we discuss below, bank regulation rations direct lending to firms, so firms borrow from the BDC rather than directly from the bank.

Liquidity shocks. So far, the firm setting is a simplified version of [Holmström and Tirole \(1998\)](#). One key departure in our model is that liquidity shocks contain both idiosyncratic

¹³A microfoundation is as follows. After the liquidity shock is drawn, the entrepreneur chooses whether to exert effort. If effort is exerted, the project succeeds for sure at $t = 2$; if the entrepreneur shirks, it succeeds only with probability $\pi < 1$ and the entrepreneur enjoys a private benefit B . Assume that it is socially beneficial to implement effort, so the entrepreneur must be compensated at least $\Delta \equiv \frac{B}{1-\pi}$, leaving only $y - \Delta$ pledgeable to outside creditors.

and aggregate components, and firms can affect their exposure through costly liquidity management. End investors in [Holmström and Tirole \(1998\)](#) are rational and break even state by state, so the private sector can only insure idiosyncratic risk. In contrast, we introduce “sleepy” depositors who value convenience from bank deposits and ultimately absorb the aggregate component of credit line drawdowns.

Motivated by the evidence in [Section 3.2](#) that BDC commitment books are concentrated and their borrower shocks are correlated, we model the downstream BDC as holding a finite portfolio of firms, so undiversified liquidity risk remains at the BDC level. The following specification allows for undiversified idiosyncratic risk under finite n and incorporates systematic correlation.

Let $Z \in \{0, 1\}$ and $\xi_i \in \{0, 1\}$ denote, respectively, the aggregate and firm i ’s idiosyncratic liquidity stress. We assume

$$Z \sim \text{Bernoulli}(z), \quad \xi_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\eta_i),$$

with Z independent of $\{\xi_i\}_{i=1}^n$. Firm i is hit by the interim liquidity shock at $t = 1$ if and only if both the aggregate and idiosyncratic shocks realize; that is, $X_i \equiv Z \cdot \xi_i$ and the probability of liquidity shock is $\mathbb{P}(X_i = 1) = z\eta_i \equiv p_i$.

Each firm chooses its idiosyncratic liquidity exposure η_i (and hence $p_i = z\eta_i$) at an increasing, convex cost $c(1 - p_i)$. This cost reflects the effort of liquidity management—for example, monitoring employees to prevent them from squandering cash. We include this margin because a central implication of our model is that firms’ liquidity management becomes inefficiently lax in equilibrium as a result of the BDC’s optimal liquidity provision.

We introduce $k = \sum_{i=1}^n X_i$ to track the number of firms hit by liquidity shocks. In equilibrium, firms are ex ante symmetric with $\eta_i = \eta$ for all i , and the distribution of k is a

two-state mixture:

$$k = \begin{cases} 0, & \text{with probability } 1 - z, \\ \text{Binomial}(n, \eta), & \text{with probability } z. \end{cases} \quad (3)$$

BDC. A business development company (BDC) lends to n firms at date 0. We model a single BDC, with many BDCs in distinct segmented markets borrowing from the same bank in the background. Its $t = 1$ liquidity need is determined by the cross-sectional distribution of firm shocks. Rather than seeking financing ex post from the spot market, which breaks even state by state, the BDC uses a committed bank credit line to pool its own liquidity risk across states. We analyze this spot-market benchmark in Section 4.3.1 for comparison.

Motivated by the empirical evidence in Section 3.2, the key difference in liquidity provision between a BDC and a conventional bank arises from the BDC's undiversified commitments and its lack of a government backstop as a nonbank. Widespread liquidity shocks are thus costly for the BDC and may even lead to its own failure. However, because the credit lines it extends are commitments made at $t = 0$, the BDC cannot discretionarily satisfy drawdown requests after shocks realize. Instead, the BDC may provide partial liquidity insurance ex ante.

The BDC rations liquidity provision by choosing the capacity of its credit line from the bank, which limits the number of borrower firms the BDC can cover and is essentially the BDC's commitment power at $t = 1$. If there are more shocks than capacity, a random subset of shocked firms is covered. Rationing this way is more efficient than excluding firms ex ante, which would forgo some diversification benefits across borrower firms.

The specific setting is as follows. At $t = 0$, each firm receives a term loan of size 1 for launching the project, and a full-sized credit line commitment of ρ from the BDC. The BDC

chooses the gross interest rate charged on firms, R_N (N for *nonbank*),¹⁴ and finances its lending through a term loan of size n plus a credit line of capacity $m\rho$ from the bank at an exogenous rate R_I (I for *intermediary*), where $m \in \{0, \dots, n\}$. Firms may alternatively borrow directly from the bank, and this outside option caps the rate R_N the BDC can charge through the firms' incentive-compatibility constraint. Since firms hit by liquidity shocks do not have sufficient pledgeable income ex post (see (1)), they default on the BDC, which may in turn trigger the BDC's default to the bank when widespread liquidity shocks realize.

In a state where k firms are hit by liquidity shocks at $t = 1$, the BDC draws

$$\rho \min\{k, m\}$$

from the bank credit line. When $k > m$, the BDC cannot fund all drawdowns: it draws $m\rho$ from its bank credit line and randomly funds m of the k shocked firms; the funded firms continue operating and repay at $t = 2$. Instead of defaulting on all credit lines, this fair lottery arrangement (as in [Bernanke and Gertler, 1989](#)) maximizes repayment from portfolio firms;¹⁵ under the linear setting, it is payoff-equivalent to pro-rata rationing if firm projects could be partially liquidated and each firm received $\frac{m\rho}{k}$ (as in [Acharya, Almeida, and Campello, 2013](#)). In practice, a BDC short of liquidity may also renegotiate or draw on parent-company support; as long as these processes involve frictions and do not recover the full promised amount, the model's qualitative implication carries through.

Conditional on k shocked firms and capacity m , the probability that a given shocked firm

¹⁴We take the BDC's market power as given, which may arise from credit market frictions and the BDC's informational advantage.

¹⁵One implementation is restructuring at $t = 1$: the BDC defaults on all credit lines, and a third party initiates restructuring that randomly injects liquidity into shocked firms to improve recovery (the BDC may still survive at $t = 2$); shocked firms without liquidity injection cannot liquidate BDC's portfolio, which is pledged as collateral for bank credit.

receives liquidity is

$$s(m, k) \equiv \mathbb{E} \left[\min \left\{ 1, \frac{m}{k} \right\} \middle| X_i = 1 \right]. \quad (4)$$

Debt obligations are due at $t = 2$. The BDC's profit at a contingent state k is then

$$\Pi_k(m, R_N) = R_N (n - k) + \min \{m, k\} (y - \Delta) - R_I [n + \rho \min \{m, k\}]. \quad (5)$$

Of the $n - k$ firms not hit by liquidity shocks, each repays R_N . In addition, $\min\{k, m\}$ shocked firms draw the credit line and repay their full pledgeable income $y - \Delta$, which falls short of $R_N(1 + \rho)$ —otherwise they could have accessed the spot market to cover the liquidity shock. On the liability side, the BDC pays R_I on both its term loan and drawdowns of the bank credit line. Instead of imposing limited liability, we assume that the BDC incurs a bankruptcy cost when its profit is negative—i.e., when it defaults on obligations to the bank. From (5), the threshold $\hat{k}$ above which the BDC defaults is

$$\hat{k} = \frac{n(R_N - R_I) + m[(y - \Delta) - R_I \rho]}{R_N}. \quad (6)$$

When $k > \hat{k}$, the BDC incurs an additional bankruptcy cost $\lambda[-\Pi_k(m, R_N)]$. Therefore, the BDC's expected payoff at $t = 0$ is

$$\Pi(m, R_N) = \mathbb{E} [\Pi_k(m, R_N)] + \lambda \sum_{k > \hat{k}} \mathbb{P}(k) \Pi_k(m, R_N) \quad (7)$$

Bank. Although banks are better positioned to provide liquidity insurance, the credit chain arises from regulatory arbitrage: banks face lower risk-weighted capital charges when lending indirectly through BDCs (Chernenko, Ialenti, and Scharfstein, 2025).

The bank is large and backed by sleepy depositors, which allow it to absorb aggregate liquidity shocks at the top of the intermediation chain. Although we model a single bank—

BDC pair for clarity, the bank holds a diversified portfolio, also lending to many other BDCs operating in distinct segmented markets. The resulting two-layer insurance structure has the bank insuring the BDC against undiversified liquidity risk, and the BDC in turn insuring borrower firms subject to frictions described above.

In our model, firms may also borrow from the bank, which serves as their outside option. For simplicity, we capture bank regulation as rationing of direct lending to these middle-market firms, who obtain bank credit only with probability $q < 1$. Because our focus is on frictions at the BDC level, we take bank rates as exogenous. Specifically, the bank charges a gross rate R_B when lending directly to firms—on both term loans and credit line drawdowns—and a gross rate R_I on credit extended to the BDC.

We now derive the bank's participation constraint at $t = 0$. Consider a state at $t = 1$ in which k borrower firms are hit by the liquidity shock. The bank advances $n + \rho \min\{k, m\}$ in total and is owed $R_I[n + \rho \min\{k, m\}]$. At $t = 2$, the BDC collects $R_N(n - k) + \min\{k, m\}(y - \Delta)$ from borrower firms, since shocked firms default and repay only their pledgeable amount $y - \Delta$. When k is large, this falls short of what the BDC owes the bank, and the BDC defaults. Hence, the actual repayment the bank collects is

$$\min \{R_I[n + \rho \min\{k, m\}], R_N(n - k) + \min\{k, m\}(y - \Delta)\}.$$

Taking expectations over k , the bank is willing to participate ex ante if lending to the BDC beats its outside option rate R ,

$$\mathbb{E} \{ \min \{R_I[n + \rho \min\{k, m\}], R_N(n - k) + \min\{k, m\}(y - \Delta)\} \} \geq R[n + \rho \mathbb{E}[\min\{k, m\}]]. \quad (8)$$

The last piece of the credit chain is depositors providing the ultimate funding to the bank. Because the bank holds a large, diversified portfolio of such credit commitments, the

idiosyncratic component of liquidity shocks washes out, leaving only the aggregate shock z . Depositors participate ex ante at $t = 0$, prior to the realization of z , and do not react to the realization of aggregate shocks because they value bank convenience and enjoy FDIC insurance, and are therefore willing to supply funds at rates that do not fully price aggregate risk ex post. By contrast, in [Holmström and Tirole \(1998\)](#), investors are fully rational and must break even upon the realization of aggregate shocks, making it impossible for the private sector to insure aggregate risk. In our setting, there is effectively a transfer from depositors to the banking sector upon aggregate shocks, and depositors serve as the ultimate funding source of liquidity.

4.2 Equilibrium Definition

Firms' problem. At $t = 0$, firms choose the effort level of liquidity management, which determines their liquidity risks via their idiosyncratic liquidity exposure. Then, they choose whether to borrow from the BDC or the bank—in equilibrium, they choose the BDC, and the bank serves as their outside option.

We analyze the firms' problem via backward induction. Given the liquidity choice of p , a firm compares payoffs from each lender. If it seeks bank financing at rate R_B , it gets financing only with probability q , but its interim liquidity shock is always covered thanks to the bank's perfect diversification and sleepy depositors. Hence, the payoff is

$$q [(1 - p) (y - R_B) + p\Delta].$$

Otherwise, if the firm borrows from the BDC, it receives only limited liquidity insurance:

conditional on being shocked, it is covered with probability

$$s(m) \equiv \mathbb{E}_k[s(m, k)] \leq 1, \quad (9)$$

where $s(m, k)$ is given in (4). Its expected payoff is

$$(1 - p)(y - R_N) + p s(m) \Delta.$$

Therefore, for a firm to choose the BDC, the following incentive-compatibility constraint holds,

$$q [(1 - p)(y - R_B) + p\Delta] \leq (1 - p)(y - R_N) + p s(m) \Delta.$$

or equivalently,

$$R_N \leq \bar{R}_{IC}(m) \equiv \frac{p\Delta (s(m) - q)}{1 - p} + (1 - q)y + qR_B. \quad (10)$$

Individual firms can affect their liquidity risks via their idiosyncratic liquidity exposure at an increasing convex cost $c(\cdot)$. Firm i chooses its liquidity shock probability p_i (equivalently, $\eta_i = \frac{p_i}{z}$) to maximize its expected payoff from borrowing from the BDC, taking the BDC's credit line capacity m and other firms' choices p_{-i} as given.

$$\max_{p_i} (1 - p_i)(y - R_N) + p_i s_i(m; p_{-i}) \Delta - c(1 - p_i),$$

where $c(\cdot)$ is increasing and convex: since a higher p_i corresponds to a larger liquidity-shock exposure, the argument $1 - p_i$ captures the firm's costly effort to reduce its liquidity risk. $s_i(m; p_{-i})$ is firm i 's coverage probability given in (9).

BDC's problem. The BDC's problem is summarized as follows. At $t = 0$, the BDC chooses the gross interest rate charged on firms, R_N , and the capacity of its credit line $m\rho$ from the bank, to maximize its expected profits $\Pi(m, R_N)$ in (7), subject to firms' incentive-compatibility constraints (10), the BDC's participation constraint that it is willing to lend to borrower firms, and the bank's participation constraint in (8).

$$\begin{aligned}
\max_{m, R_N} \quad & \Pi(m, R_N) = \mathbb{E}[\Pi_k(m, R_N)] + \lambda \sum_{k > \hat{k}} \mathbb{P}(k) \Pi_k(m, R_N) \\
\text{s.t.} \quad & q[(1-p)(y - R_B) + p\Delta] \leq (1-p)(y - R_N) + p s(m) \Delta, & (\text{Firm IC}) \\
& R_N \leq y - \Delta, & (\text{Pledgeability}) \\
& R_N(1-p) + p s(m)(y - \Delta) \geq R_I [1 + p s(m)\rho], & (\text{BDC PC}) \\
& \mathbb{E}[\min\{R_I[n + \rho \min\{k, m\}], R_N(n - k) + \min\{k, m\}(y - \Delta)\}] \\
& \geq R[n + \rho \mathbb{E}[\min\{k, m\}]]. & (\text{Bank PC})
\end{aligned}$$

As discussed, we verify the bank's participation constraint ex post. It holds if the bank charges a sufficiently high rate R_I . For a fixed credit line size $m\rho$, BDC value is linearly increasing in R_N , so it optimally chooses the maximum feasible rate. The firm IC constraint imposes an upper bound on R_N to make the BDC option attractive, and the firm cannot repay more than its pledgeable income $y - \Delta$. The BDC's participation constraint, on the other hand, imposes a lower bound on R_N to make lending to borrower firms profitable. To summarize, the BDC chooses:

$$R_N^*(m) = \min\{y - \Delta, \bar{R}_{IC}(m)\}. \quad (11)$$

Therefore, the BDC chooses the optimal capacity to maximize

$$\max_{m \in \{0, \dots, n\}} \Pi^*(R_N^*(m), m).$$

4.3 Model Implications

We first discuss a benchmark without bank credit lines, in which the BDC raises liquidity from the financial market ex post, to isolate the insurance it can provide on its own. We then characterize the equilibrium in the credit chain and show that the BDC optimally provides only partial insurance given its undiversified portfolio risk. This generates a novel implication for nonbank credit provision: borrower firms inefficiently shirk in liquidity management, an externality that arises from their sharing of a constrained liquidity pool.

4.3.1 Liquidity Provision Without Bank Credit Lines

For idiosyncratic liquidity shocks, any intermediary with commitment power—not necessarily a bank—can provide liquidity insurance, in essence the idea of [Diamond \(1984\)](#). The BDC is precisely such an intermediary, so it can provide some insurance on its own even without a bank. We isolate this channel by assuming the BDC raises liquidity from the financial market at $t = 1$ while holding all other assumptions fixed. The key difference between the market and a bank credit line is that the market does not commit and must break even state by state.

The limit $n \rightarrow \infty$ gives the sharpest illustration of diversification. Idiosyncratic risk is fully diversified, so the market responds to the realized aggregate state Z . When $Z = 0$, no firm is shocked and no funding is needed; when $Z = 1$, a fraction η of firms is shocked. The

market's break-even condition in the $Z = 1$ state is satisfied when η is small,

$$R_N(1 - \eta) + \eta(y - \Delta) \geq R_I(1 + \eta\rho).$$

In this case, the BDC fully insures its portfolio on its own, as the bank could.¹⁶

More generally, consider the finite- n case of our model. When k borrower firms are hit, the BDC raises funding from the market at $t = 1$ via senior claims; existing creditors who fund loans at $t = 0$ are willing to be diluted to allow firms to continue. A draw of ρk can be funded if and only if the BDC's pledgeable income at $t = 2$ is sufficient to cover repayment at a gross rate of R_I ,

$$R_N(n - k) + k(y - \Delta - \rho R_I) \geq R_I n, \quad (12)$$

where R_N is capped by firms' incentive-compatibility constraint relative to their outside option of borrowing from the bank directly. The left-hand side decreases in k because lending to shocked firms is negative-NPV, as $y - \Delta < \rho R_I$. The condition also requires the BDC to charge a sufficiently high rate, $R_N \geq R_I$, so that repayments from unshocked firms can cross-subsidize the shocked ones. Then (12) defines a unique threshold $k_0(R_N)$ such that liquidity shocks can be fully funded from the market ex post if and only if $k \leq k_0$.

Otherwise, when more borrower firms are hit, the BDC can at best partially cover $l(k) < k$ of the shocked firms, and the remaining shocked firms are liquidated. The BDC covers $l(k)$ firms whenever its pledgeable cash flows from borrower firms satisfy

$$R_N(n - k) + l(k)(y - \Delta) \geq R_I[n + \rho l(k)].$$

¹⁶When η is large, the inequality fails, and the BDC can provide only partial insurance by rationing. This leaves an essential role for the bank, whose sleepy depositors free it from having to break even state by state given Z .

Therefore, the BDC provides partial insurance from portfolio diversification even without a bank credit line, whereas individual firms cannot obtain such insurance from the market on their own, as discussed around (1). Because market funding responds to the BDC's portfolio realizations, firms expect worse coverage when more shocks occur. Relatedly, because the market only lends when the BDC can fully repay, this disciplinary force also reduces the chance that the BDC defaults from overextending credit. As we show in the next section, a committed bank credit line may instead cause the BDC to default upon widespread portfolio shocks, which is a key reason the BDC rations liquidity provision by choosing a small credit line capacity.

4.3.2 Liquidity Provision in the Credit Chain

We have shown that the BDC provides a layer of liquidity insurance by pooling liquidity risk across its borrower firms (Section 4.3.1). Undiversified liquidity shocks nonetheless remain, owing to the BDC's small portfolio size and the concentration of its commitment book documented in Section 3.2. These residual shocks constitute the BDC's own liquidity risk. In the credit chain, the committed bank credit line insures the BDC across these states, forming the second layer of insurance.

In this part, we show that even with access to a bank credit line, the BDC may optimally provide only *partial liquidity insurance*, essentially because of its undiversified borrower liquidity risk. When widespread shocks materialize, full coverage would force the BDC to default and incur bankruptcy costs. The BDC limits this exposure by rationing coverage ex post, which it implements through taking a smaller bank credit line, as we show next in Proposition 1.

Proposition 1. *There exists a cutoff bankruptcy cost $\hat{\lambda} > 0$ such that*

- (i) *If $\lambda > \hat{\lambda}$, the BDC selects a small credit line from the bank that rations borrower-firm*

drawdowns in widespread-shock states and the liquidity provision is partial:

$$1 \leq m^* < \hat{k};$$

(ii) If $\lambda < \hat{\lambda}$, the credit line from the bank fully covers all potential borrower-firm drawdowns:

$$m^* = n.$$

Figure 6 illustrates the BDC's value as a function of the bank credit line capacity m under two scenarios. When bankruptcy costs are high, the BDC chooses an interior m^* that is smaller than the threshold $\hat{k}$ that leads to its own insolvency. When bankruptcy costs are low, the BDC chooses full coverage with $m^* = n$.

Essentially, bank credit-line capacity m determines how much of the BDC's borrower liquidity risk can be covered ex post; by choosing m , the BDC sets its commitment power for states with widespread portfolio shocks. Because firms receive a constant outside-option payoff and the bank earns a fixed unit spread on extended credit, the social benefit and cost of increasing m both accrue to the BDC.

Marginally increasing the credit line from the bank only matters in ex post states of widespread liquidity shocks, where the bank credit line binds. A larger line allows more of the BDC's shocked borrower firms to continue, reducing inefficient liquidation. This benefit accrues to the BDC through a higher rate R_N that it can charge on borrower firms: with greater capacity, firms obtain higher coverage from the BDC and are therefore willing to pay higher rates. However, ex post liquidity injections are socially costly because part of firm output is not pledgeable. Beyond these social benefits and costs, a third force operates purely at the BDC level: in states with many drawdowns, covering additional firms erodes the BDC's own solvency and raises its expected bankruptcy cost.

Therefore, given that the BDC cannot fully diversify borrower liquidity risk, it faces a tradeoff in expanding coverage against widespread drawdowns, and optimally rations liquidity insurance when bankruptcy costs are high. This tradeoff generates the threshold in Proposition 1. When bankruptcy costs are low, the BDC finds it worthwhile to insure every borrower firm. When bankruptcy costs are high, the BDC chooses limited capacity and accepts that some firms go uncovered in systemic stress states.

This mechanism maps to our empirical findings. The premise—that the BDC’s liquidity risk from its commitment book is undiversified—is documented in Section 3.2, where the book’s concentration and the commonality of borrower shocks leave the BDC with potentially large residual exposure. The prediction of partial coverage is consistent with the levels evidence in Table 9: each dollar of unfunded commitment is matched by roughly \$0.53 of *undrawn* bank credit capacity, short of the dollar-for-dollar coverage that full insurance ($m^* = n$) would require. This gap is in fact a conservative reading of the model. In practice, BDCs also draw on their bank credit lines to fund other liquidity needs—most notably new investment—rather than commitment drawdowns alone (Section 3.3.2 and Table IA.2). The \$0.53 of undrawn capacity per dollar of commitment is therefore an *upper bound* on the slack actually earmarked for covering borrower-firm drawdowns, reinforcing the interpretation that liquidity insurance is partial.

4.3.3 Firms’ Liquidity Management

A unique implication of the BDC’s partial liquidity insurance provision is that firms *inefficiently* shirk in liquidity management given a shared, constrained liquidity pool. The core idea is a *tragedy of the commons* in the spirit of Hardin (1968).

When an individual firm chooses its effort level for liquidity management, it only considers how its exposure to liquidity shock would affect its own performance and liquidity need.

Formally, the individual firm's first-order condition in the symmetric equilibrium is

$$-c'(1 - p^{eq}) = -(y - R_N) + \Delta s(m), \quad (13)$$

Condition (13) balances the marginal benefit and cost of a higher exposure p^{eq} from lax liquidity management. Shirking saves the effort cost $c'(1 - p^{eq})$ but raises the chance of an interim shock; that shock reduces the firm's $t = 2$ payoff, against which it is covered by the BDC credit line only with probability $s(m) \leq 1$.

However, the firm considers only its own chance of drawing on the credit line and ignores that doing so crowds out coverage for the BDC's other borrower firms, which share the same liquidity pool. To see this externality, a planner instead chooses the exposure p to maximize all firms' joint payoff:

$$\max_p \sum_i \left[(1 - p_i)(y - R_N) + p_i s_i(m; p_{-i}) \Delta - c(1 - p_i) \right].$$

This internalizes how each firm's liquidity management affects coverage for the rest of the pool. Relative to the individual firm's problem, the planner's first-order condition adds a congestion wedge,

$$-c'(1 - p^{planner}) = -(y - R_N) + \Delta s(m) + \underbrace{\Delta (n - 1) \eta \frac{\partial s_j}{\partial \eta_i} \Big|_{\text{sym}}}_{\leq 0}.$$

A higher exposure by firm i raises the chance that the pool's capacity binds and lowers every other firm's coverage, so $\partial s_j / \partial \eta_i \leq 0$, strictly whenever $m < n$ (Appendix A). Because the individual firm ignores this externality, the non-positive wedge—vanishing only when coverage never binds—implies that firms inefficiently exert less effort in liquidity management

relative to the socially optimal level. The following proposition summarizes the result.

Proposition 2. *Suppose $\lambda > \hat{\lambda}$, so $s(m^*) < 1$. Then firms' equilibrium liquidity exposure exceeds the planner's choice: $p^{eq} > p^{planner}$. When $\lambda < \hat{\lambda}$, $s(m^*) = 1$ and the two coincide.*

When $\lambda > \hat{\lambda}$, Proposition 1 implies $m^* < \hat{k}$ and hence $s(m^*) < 1$: liquidity insurance is *constrained*. The BDC's constrained credit-line capacity is then a common liquidity pool that its borrower firms draw on in widespread-shock states, and it is precisely this rationing that turns each firm's private liquidity choice into an externality on the others.

This means that the externality is specific to nonbanks that may provide constrained liquidity insurance. We have shown that BDCs face undiversified liquidity shocks because their portfolios are concentrated and their borrowers' risks are correlated, and that, lacking government backstops, they may optimally ration liquidity provision to preempt the bankruptcy costs of widespread shocks. All of this yields *constrained* liquidity provision that is unique to nonbanks.

The mechanism echoes the simultaneous drawdowns of committed bank lines studied by [Bräuning and Ivashina \(2026\)](#), when firms draw on their facilities at once and strain the capacity of their lenders. The difference is that theirs is an *ex-post* run: borrowers rationally exercise options on committed capacity once an aggregate shock hits, and the resulting strain is a realization of risk that was already in place. Ours is an *ex-ante* externality: because the shared pool will be rationed in widespread-shock states, firms anticipate that their own draws will compete with others' and respond by exerting less effort than socially optimal in liquidity management. The ex-post run channel is absent in our model because of timing—drawdowns occur only after liquidity shocks are realized, so a firm not hit gains nothing from drawing and must still repay interest. Firms cannot run away from repaying either: liquidity shocks are verifiable; and were the BDC to fail, the bank would collect the firms' repayment instead—in our data, BDCs' bank revolving facilities are predominantly senior secured by

the underlying portfolio.

5 Conclusion

As credit extension migrates from banks to nonbanks, a basic question is where the boundary of banking lies—what banks can do that nonbanks cannot. Liquidity insurance, the commitment to advance cash when a borrower is hit by a shock, has long been the province of deposit-funded banks. We document a new and economically important fact: business development companies (BDCs), as nonbank lenders, now supply it on a significant scale.

We establish three empirical patterns. BDCs’ credit commitments have grown large—with commitment-to-asset ratios comparable to that of banks—and continue to rise. BDCs’ commitment book is concentrated and its borrowers’ draws are correlated, leaving undiversified liquidity risks at the BDC level. BDCs cover this residual risk with committed credit lines from banks as a liquidity backstop.

We rationalize these patterns with a model of a credit chain in which liquidity insurance is provided in two layers. The BDC forms the first layer, partially pooling the idiosyncratic shocks of its borrower firms; the bank forms the second, committing liquidity to the BDC against the undiversified risk that remains, with its sleepy depositors as the ultimate source of liquidity. The model delivers two implications unique to nonbank liquidity provision. First, because of its undiversified borrower liquidity shocks and its lack of a public backstop, the BDC may optimally provide only partial liquidity insurance. Second, this constrained provision yields a novel externality in firms’ liquidity management: each firm exerts inefficiently little effort because it ignores how its own draw on the shared pool crowds out the others—a tragedy of the commons specific to nonbanks’ constrained liquidity provision.

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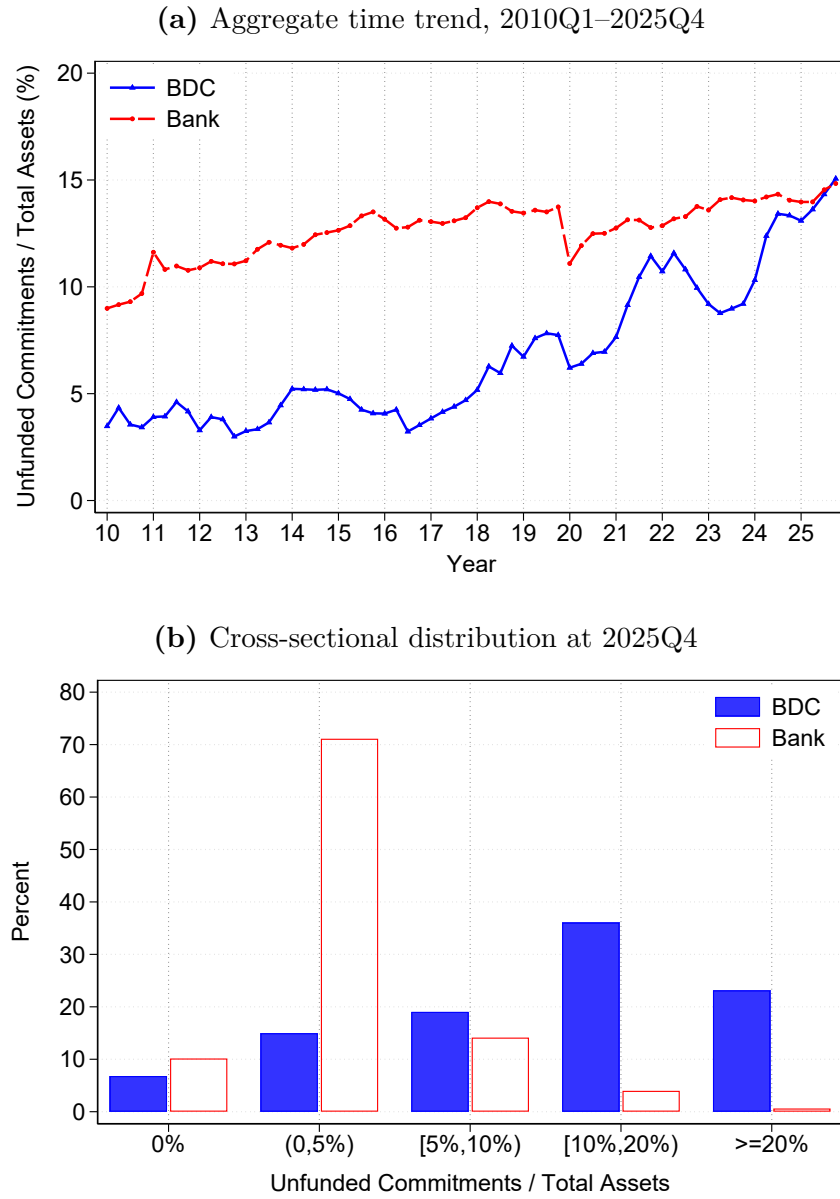


Figure 1. Liquidity Provided by Intermediaries: BDCs vs Banks.

This figure shows liquidity insurance provided by BDCs and banks to their corporate borrowers, measured as unfunded commitments divided by total assets. For BDCs, the numerator is total unfunded commitments. For banks, the numerator is the sum of “unused commitments” (Call Report Schedule RC-L) to commercial and industrial (C&I) borrowers and to financial institutions. Panel (a) plots quarterly aggregate ratios across all banks and all BDCs from 2010Q1 to 2025Q4. Panel (b) presents the cross-sectional distribution at 2025Q4. Data sources: SEC’s EDGAR and FFIEC’s Consolidated Reports of Condition and Income.

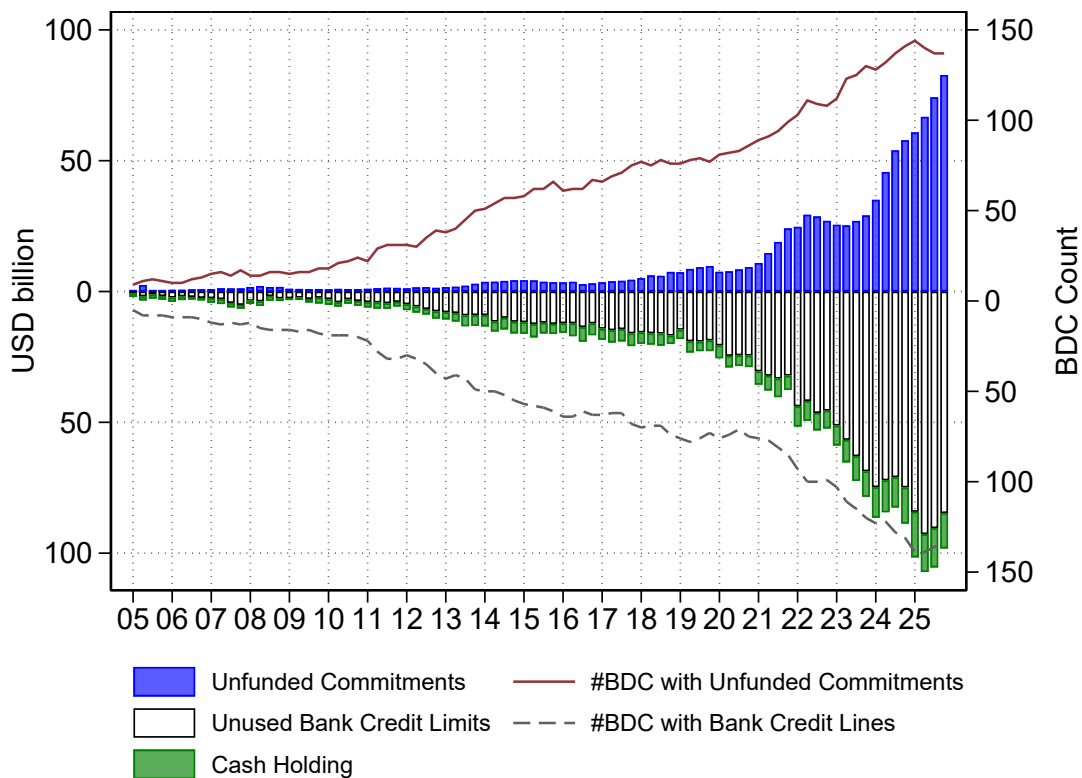


Figure 2. Liquidity Provided by and Available to BDCs.

This figure presents quarterly aggregate amounts of unfunded commitments provided by BDCs—undrawn by portfolio companies—and the undrawn commitment available to them between 2005Q1 and 2025Q4. The blue bars show the dollar amount of unfunded loan commitments extended by BDCs to their portfolio companies. The white bars show the dollar amounts of undrawn credit limits on the bank credit lines BDCs receive as borrowers. The green bars show BDCs' total cash holdings. Solid red and dashed grey lines indicate the number of BDCs that have unfunded loan commitments and undrawn bank credit lines, respectively. Data sources: SEC EDGAR, Compustat, and S&P Capital IQ.

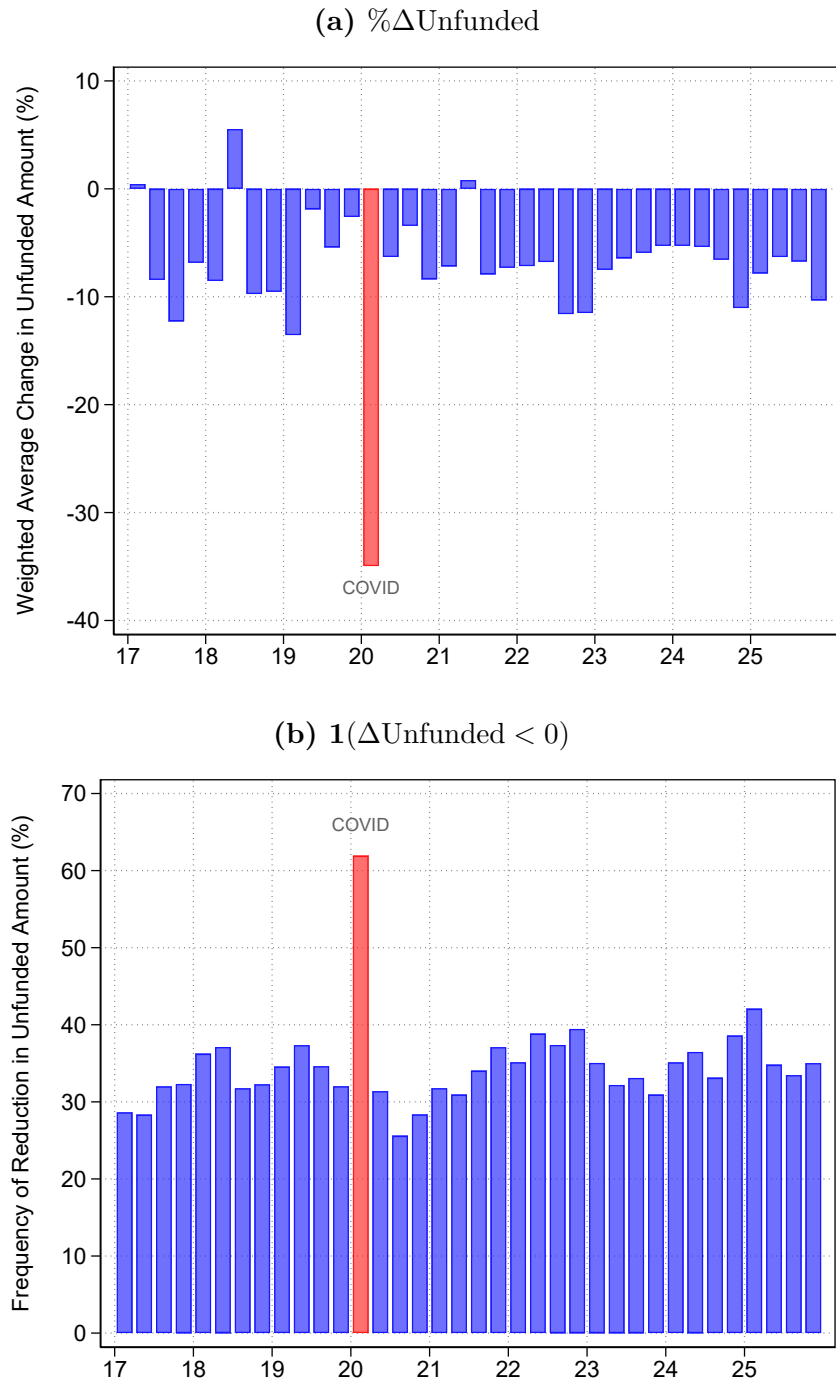


Figure 3. Changes in Unfunded Commitments.

This figure presents the time series of changes in unfunded commitments extended by BDCs to their borrowers from 2017Q1 to 2025Q4. Panel (a) shows the average quarterly percentage change in unfunded commitment amounts. Panel (b) displays the quarterly frequency of reductions in unfunded commitment amounts. The red bar highlights 2020Q1 corresponding to the onset of the COVID-19 pandemic.

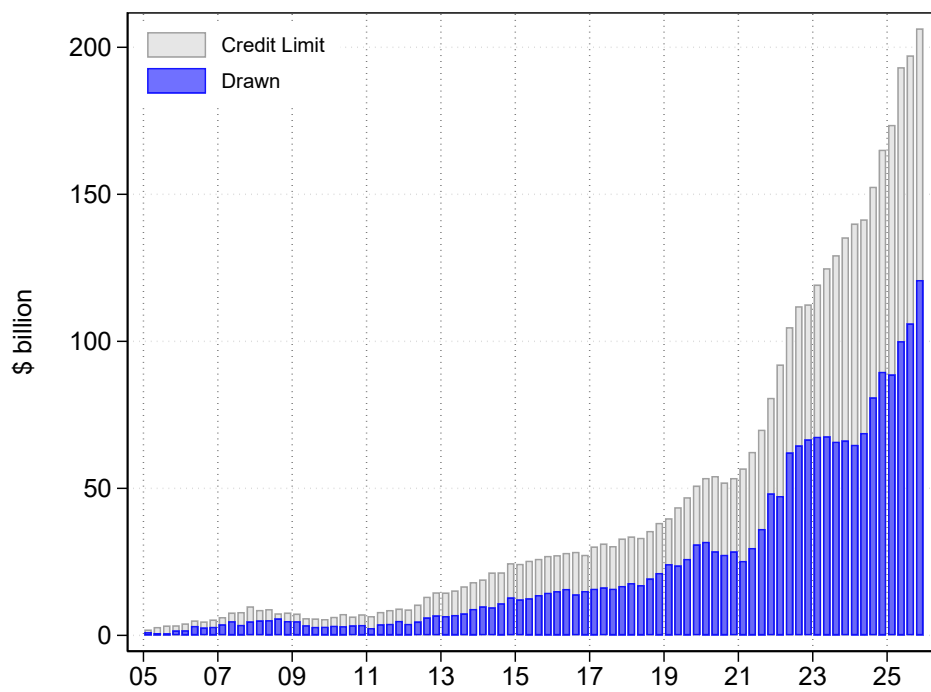


Figure 4. BDC Credit Lines from Banks: Credit Limits and Drawn Amounts. This figure plots aggregate bank credit line capacity (gray) and drawn amounts (blue) across all BDCs by quarter from 2005Q1 to 2025Q4. Credit lines are revolving facilities that BDCs obtain from banks to fund their lending and investment activities. The gap between the two bars represents the undrawn credit limit available to BDCs. Both total capacity and drawn amounts have grown steadily over the sample, reflecting the expansion of the BDC sector.

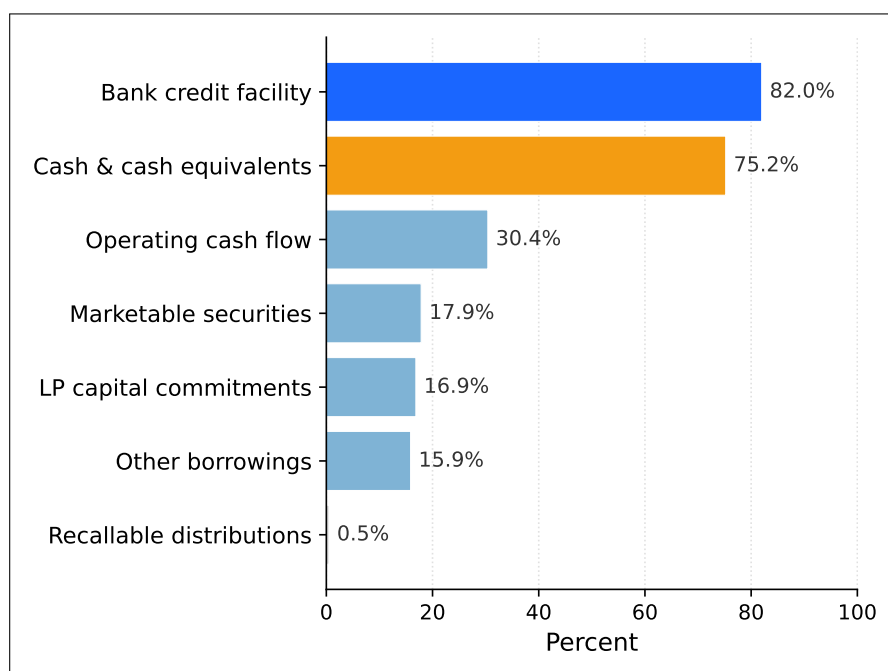


Figure 5. Liquidity Sources Disclosed as Backstop for Unfunded Commitments.

This figure reports the share of BDC 10-K filings in which the BDC discusses specific sources of liquidity as backstop for its unfunded commitments to portfolio companies. The sample is restricted to the 605 10-K filings in which the BDC explicitly names at least one specific source of liquidity as backstop for unfunded commitments. Internet Appendix Figure [IA.2](#) reports the time-series of disclosure status across the full 2006–2025 sample.

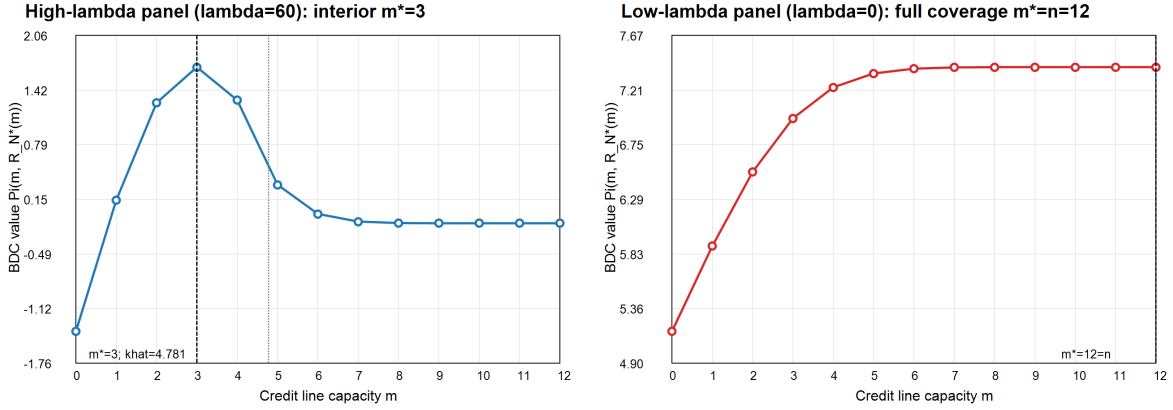


Figure 6. Optimal Size of Credit Line from Banks. The figure plots the BDC value $\Pi(m, R_N^*(m))$ as a function of the commitment capacity of the bank credit line m . The left panel corresponds to the high- λ case, which delivers an interior m^* (dashed vertical line) that is smaller than the threshold number of shocks that leads to BDC bankruptcy, $\hat{k}(m^*)$ (dotted vertical line). The right panel corresponds to the low- λ case, which delivers full coverage for borrower firms $m^* = n$. Parameters: $n = 12$, $z = 0.45$, $\eta = 0.25$, $q = 0.75$, $y = 4.00$, $\Delta = 2.00$, $R_I = 1.02$, $\rho = 2.30$, $R_B = 1.10$, and $R = 1.00$, with the success probability normalized to one. The high- λ panel uses $\lambda = 60$, and the low- λ panel uses $\lambda = 0$.

Table 1. Summary of BDC Sample

This table summarizes the BDC-quarter panel sample. Panel A reports all BDCs between 2005Q1 and 2025Q4 with non-missing balance-sheet items (total assets, total debt, equity, and cash and equivalents). *Total Assets*, *Total Debt*, *Equity*, and *Cash* are the BDC’s balance-sheet items. *Total Unfunded* is the total amount of unfunded commitments the BDC extends to its borrowers. *N Bank Credit Lines* is the number of revolving credit lines the BDC obtains from its lenders, and *Bank Credit Limit* and *Undrawn Bank Credit* are the corresponding aggregate and unused capacity. *Inv. Cash Outflow* is the BDC’s quarterly gross investment cash outflow. Panel B reports a subsample of BDC-quarters for which details on the BDC’s commitments extended to its portfolio companies are reported in regulatory filings. *N Unfunded* is the number of individual portfolio commitments. *Mean Unfunded* and *Median Unfunded* are the within-BDC-quarter average and median amounts of undrawn credit commitment to portfolio companies in \$ millions. Panel C reports the fair-value share of asset classes in BDC portfolios. Senior Secured Loans includes first/second lien term loans, unitranche, and the drawn portion of senior secured revolvers and DDTLs. Joint Ventures captures equity-style interests in BDC-managed joint-venture lending vehicles whose underlying economic exposure is primarily senior secured loans. Equity includes the remaining common, preferred, warrants, and partnership/LLC interests in operating portfolio companies. Other Debt includes subordinated, mezzanine, unsecured, and convertible debt as well as structured-finance and miscellaneous positions. All amounts are measured in USD million.

	N	mean	sd	p5	p25	p50	p75	p95
<i>Panel A: All BDC-Quarters</i>								
Total Assets	6,746	1,519.4	4,088.1	1.4	147.7	483.7	1,385.6	5,807.5
Total Debt	6,746	662.6	1,926.2	0.0	21.0	182.6	603.3	2,283.7
Equity	6,746	811.3	2,154.9	0.2	96.3	268.7	744.1	3,181.0
Cash	6,746	57.0	137.3	0.0	4.4	18.1	53.9	233.8
Total Unfunded	6,747	147.8	595.3	0.0	0.2	19.5	109.5	531.0
N Bank Credit Lines	6,747	1.4	1.6	0.0	0.0	1.0	2.0	4.0
Bank Credit Limit	6,747	566.4	1,454.1	0.0	0.0	175.0	597.5	1,975.0
Undrawn Bank Credit	6,747	261.3	690.6	0.0	0.0	67.5	245.7	1,080.4
Inv. Cash Outflow	5,911	234.3	853.0	0.3	20.7	68.2	193.8	853.0
<i>Panel B: BDC-Quarters with Reported Unfunded Commitment Book</i>								
N Commitments	2,977	44.5	56.2	2.0	8.0	22.0	60.0	165.0
Mean Unfunded	2,977	4.9	18.5	0.4	1.3	2.6	5.1	14.2
Median Unfunded	2,977	3.4	17.7	0.3	0.8	1.8	3.4	9.9
<i>Panel C: BDC-Quarter with Portfolio Holdings</i>								
Senior Secured Loans	5,811	73.6%	31.9%	0.0%	65.8%	87.4%	96.9%	100.0%
Joint Ventures	5,811	0.8%	3.0%	0.0%	0.0%	0.0%	0.0%	5.5%
Equity	5,811	11.8%	20.2%	0.0%	1.0%	4.8%	12.7%	59.5%
Other Debt	5,811	13.7%	23.7%	0.0%	0.0%	2.4%	14.5%	77.3%

Table 2. Summary of Unfunded Commitment Book Sample

This table summarizes the sample of credit commitments extended by BDCs to their individual portfolio companies between 2005Q1 and 2025Q4, at the BDC–borrower–commitment-type–quarter level. The sample requires the same borrower-commitment type to be observed in both the current and prior quarter and lagged unfunded commitment above \$1 million. This filter removes originations, terminations, and small-denominator observations for which percentage changes are mechanically noisy. Panel A reports summary statistics for all commitment types. Panel B and Panel C report the corresponding statistics for revolvers and DDTLs. All amounts are measured in USD million. *Unfunded* is the amount of undrawn credit commitment. $\Delta Unfunded$ and $\% \Delta Unfunded$ are the change and percentage change in the unfunded amount from the previous period, respectively. $\mathbf{1}(\Delta Unfunded < 0)$ is an indicator variable that equals one if the unfunded amount decreases, and zero otherwise. *Time to Maturity* is the remaining time to facility expiration in years, computed as the number of years between the expiration date and the report date (available for the subsample of commitments where the filing reports an expiration date).

	N	mean	sd	p5	p25	p50	p75	p95
<i>Panel A: All Commitment Types</i>								
Unfunded	66,325	7.021	17.128	0.983	1.741	3.214	6.910	23.800
$\Delta Unfunded$	66,325	-0.413	6.817	-4.150	-0.359	0.000	0.000	1.388
$\% \Delta Unfunded$	66,325	-0.024	0.421	-0.597	-0.101	0.000	0.000	0.396
$\mathbf{1}(\Delta Unfunded < 0)$	66,325	0.345	0.475	0.000	0.000	0.000	1.000	1.000
Time to Maturity	17,973	3.268	1.921	0.367	1.454	3.526	4.920	6.174
<i>Panel B: Revolvers</i>								
Unfunded	21,773	5.035	10.141	1.000	1.528	2.500	5.000	16.250
$\Delta Unfunded$	21,773	-0.102	3.627	-1.883	-0.002	0.000	0.000	1.119
$\% \Delta Unfunded$	21,773	-0.003	0.347	-0.474	0.000	0.000	0.000	0.379
$\mathbf{1}(\Delta Unfunded < 0)$	21,773	0.260	0.439	0.000	0.000	0.000	1.000	1.000
Time to Maturity	7,484	4.207	1.351	1.711	3.335	4.372	5.251	6.251
<i>Panel C: Delayed Draw Term Loans</i>								
Unfunded	16,717	8.369	16.454	0.906	1.820	3.629	7.945	31.095
$\Delta Unfunded$	16,717	-0.674	6.963	-5.756	-0.660	0.000	0.000	0.523
$\% \Delta Unfunded$	16,717	-0.054	0.443	-0.651	-0.167	0.000	0.000	0.100
$\mathbf{1}(\Delta Unfunded < 0)$	16,717	0.415	0.493	0.000	0.000	0.000	1.000	1.000
Time to Maturity	7,567	2.340	1.947	0.249	0.903	1.506	4.207	6.387

Table 3. BDC-Level Liquidity Shocks from Commitment Books

This table reports BDC-level volatility and magnitude of liquidity shocks from the commitment book. Panel A compares the volatility of shocks against a diversified benchmark. The benchmark equal-weights all unfunded commitments extended to borrowers in the BDC universe. The sample consists of 59 BDCs with at least 12 quarters between 2005Q1 and 2025Q4 in which the BDC has at least 5 continuing borrowers in its commitment book. Two shock measures are reported. The continuous shock is the percentage change in unfunded commitment, $\Delta u_{i,\ell,t}/u_{i,\ell,t-1}$; the binary shock is the indicator $\mathbf{1}\{\Delta u_{i,\ell,t} < 0\}$. For each BDC ℓ , μ_ℓ and σ_ℓ are the time-series mean and standard deviation of BDC ℓ 's own shock series (dollar-weighted across its continuing borrowers in the continuous case, equal-weighted in the binary case). σ_ℓ^{BMK} is the time-series standard deviation of the diversified-benchmark shock computed over the same quarters as BDC ℓ . Panel B reports the estimated magnitude of liquidity shocks, using 2,258 BDC-quarters of the same 59 BDCs. Dollar-valued liquidity shock associated with one-standard-deviation quarterly change in the unfunded commitments is estimated as $\hat{\sigma}_{\ell t}^{\$} \equiv \sigma_\ell \times \text{Total Unfunded}_{\ell,t-1}$, where σ_ℓ is the BDC's own volatility from Panel A, and $\text{Total Unfunded}_{\ell,t-1}$ is its lagged total unfunded commitments. Rows 2-4 split the sample into terciles by the ratio of Total Unfunded to Total Assets. $\mathbf{1}(\hat{\sigma}_{\ell t}^{\$} > \text{Cash})$ is an indicator for whether the estimated shock exceeds lagged cash.

Panel A: Volatility of Liquidity Shocks

	$\Delta \text{Unfunded}$				$\mathbf{1}(\Delta \text{Unfunded} < 0)$			
	μ_ℓ	σ_ℓ	σ_ℓ^{BMK}	$\sigma_\ell/\sigma_\ell^{\text{BMK}}$	μ_ℓ	σ_ℓ	σ_ℓ^{BMK}	$\sigma_\ell/\sigma_\ell^{\text{BMK}}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
mean	-0.072	0.145	0.048	3.582	0.367	0.142	0.057	2.765
p25	-0.105	0.090	0.030	1.760	0.315	0.123	0.037	1.807
p50	-0.073	0.116	0.054	2.587	0.371	0.137	0.065	2.461
p75	-0.047	0.181	0.063	4.415	0.426	0.158	0.074	3.425

Panel B: Magnitude of Liquidity Shocks

	N	Median Total	Median	Mean
	(1)	Unfunded/TA	$\hat{\sigma}_{\ell t}^{\$}/\text{Cash}$	$\mathbf{1}(\hat{\sigma}_{\ell t}^{\$} > \text{Cash})$
	(1)	(2)	(3)	(4)
All BDC-quarters	2,258	0.072	0.291	0.206
Total Unfunded / TA: Low	753	0.024	0.075	0.061
Total Unfunded / TA: Mid	752	0.072	0.361	0.214
Total Unfunded / TA: High	753	0.139	0.605	0.344

Table 4. Concentration of BDCs' Commitment Books

This table compares the concentration of BDCs' unfunded commitments to the concentration of their debt-only portfolio holdings. The sample consists of 2,619 BDC-quarters with data on both the commitment book and portfolio holdings. Industry, HQ-state, and sponsor measures are computed on this common set of BDC-quarters. For each BDC-quarter, concentration is summarized at three levels of aggregation (borrower industry group, HQ state, and sponsor) using two measures: the Herfindahl–Hirschman index $\sum_i w_i^2$, where w_i is the entity's share of total unfunded commitment (or fair-value of debt holdings); and the Top-3 share, the sum of the three largest weights. Each row reports the cross-sectional mean and quartiles across BDC-quarters, separately under commitment book weights (columns 1–4) and loan portfolio weights (columns 5–8). The Diff column reports the mean within-BDC-quarter difference between the unfunded and holdings values, with significance from a paired t -test, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

	Commitment Book				Loan Portfolio				Diff
	mean	p25	p50	p75	mean	p25	p50	p75	(paired- t)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>HHI</i>									
Industry	0.285	0.160	0.225	0.348	0.162	0.104	0.130	0.184	0.124***
HQ State	0.241	0.094	0.181	0.309	0.102	0.063	0.077	0.109	0.140***
Sponsor	0.220	0.061	0.129	0.295	0.061	0.025	0.037	0.067	0.159***
<i>Top-3 Share</i>									
Industry	0.730	0.594	0.715	0.860	0.562	0.465	0.533	0.639	0.168***
HQ State	0.631	0.421	0.625	0.813	0.416	0.321	0.373	0.466	0.215***
Sponsor	0.563	0.321	0.515	0.809	0.272	0.169	0.219	0.332	0.291***
N	2,619								

Table 5. BDC Specialization in Extending Commitments

This table measures BDCs' specialization in unfunded commitments relative to the cross-BDC aggregate commitment book, following Blickle, Parlatore, and Saunders (2025). The specialization measure is the *excess share*: how much a BDC's weight in a given industry, state, or sponsor exceeds the cross-BDC aggregate weight in the same dimension. Top-1 picks the BDC's most over-weighted bucket within the BDC-quarter; Top-3 sums the weights of the three largest over-weighted buckets. The sample consists of 2,619 BDC-quarters with data on both the commitment book and portfolio holdings. For each BDC-quarter and each industry-group, HQ-state, or sponsor bucket, the excess share is the BDC's own weight in the bucket minus the aggregate weight across BDCs in the same bucket and quarter. Each row reports the cross-sectional distribution of the measure, separately for commitment book weights (columns 1–4) and loan portfolio weights (columns 5–8). The Diff column reports the average within-BDC-quarter difference between the commitment book and loan portfolio values, with significance from a paired *t*-test, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

	Commitment Book				Loan Portfolio				Diff
	mean	p25	p50	p75	mean	p25	p50	p75	(paired- <i>t</i>)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Industry Excess Share</i>									
Top-1	0.292	0.159	0.246	0.379	0.157	0.080	0.121	0.204	0.135***
Top-3	0.467	0.325	0.447	0.591	0.281	0.166	0.248	0.364	0.185***
<i>HQ State Excess Share</i>									
Top-1	0.267	0.119	0.214	0.356	0.106	0.051	0.076	0.120	0.161***
Top-3	0.444	0.258	0.415	0.591	0.216	0.123	0.170	0.255	0.228***
<i>Sponsor Excess Share</i>									
Top-1	0.298	0.120	0.219	0.410	0.121	0.055	0.086	0.148	0.176***
Top-3	0.530	0.284	0.484	0.779	0.247	0.136	0.195	0.311	0.282***
N	2,619								

Table 6. Selection of Portfolio Borrowers into Commitment Books

This table reports linear probability regressions of whether a BDC's portfolio company carries an unfunded commitment on borrower characteristics and BDC-borrower relationship characteristics. The sample consists of BDC-borrower-quarter observations for borrowers in BDC portfolio holdings between 2005Q1 and 2025Q4. In column 1, the dependent variable is an indicator for whether the BDC extends any unfunded commitment to the borrower in the quarter. In columns 2 and 3, the sample restricts to BDC-quarters that report the type of commitments, and the dependent variables are indicators for whether the BDC extends a revolver-type or DDTL-type commitment, respectively. The regressors include borrower-level or borrower-quarter variables: Credit Quality, Sponsor Backed, M&A Deal, $\log(\text{Employees})$, and $\log(\text{Age})$; and BDC-borrower-quarter relationship variables: Dual Held and $\log(\text{Principal})$. Credit Quality, Dual Held, and $\log(\text{Principal})$ are measured at $t - 1$; M&A Deal equals one if the borrower has an M&A deal during quarters $t - 1$ through $t - 4$. All specifications include industry-time, HQ state-time, and BDC-time fixed effects (Time = year-quarter throughout). Standard errors are two-way clustered at the BDC and borrower levels, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

Dependent Variable: 1{Commitment Extended}			
	Any Commitment (1)	Revolver (2)	DDTL (3)
Credit Quality	0.258*** (0.034)	0.143*** (0.027)	0.062*** (0.016)
Sponsor Backed	0.036*** (0.012)	0.025** (0.010)	0.006 (0.006)
M&A Deal	0.047*** (0.007)	0.023*** (0.005)	0.031*** (0.006)
Dual Held	0.022* (0.012)	0.026** (0.011)	0.001 (0.005)
$\log(\text{Employees})$	-0.022*** (0.003)	-0.013*** (0.002)	-0.006*** (0.002)
$\log(\text{Age})$	-0.013*** (0.004)	-0.001 (0.003)	-0.008*** (0.002)
$\log(\text{Principal})$	0.020*** (0.005)	0.016*** (0.004)	0.004 (0.004)
Industry-Time FEs	Y	Y	Y
HQ State-Time FEs	Y	Y	Y
BDC-Time FEs	Y	Y	Y
N	308,429	281,743	281,743
R^2	0.402	0.413	0.257

Table 7. Commonality in Liquidity Shocks from Commitment Books

This table reports estimated coefficients from univariate panel regressions of borrower-level unfunded-commitment shocks on observable factors. The sample consists of BDC-borrower-quarters in continuing unfunded facilities (excluding originations and terminations) with $u_{i,\ell,t-1} > \$1$ million between 2005Q1 and 2025Q4 for which all eight factor series are available. The outcome is either the continuous shock $\Delta u_{i,\ell,t}/u_{i,\ell,t-1}$ or the binary shock $\mathbf{1}\{\Delta u_{i,\ell,t} < 0\}$. Panel A uses quarterly average credit line drawdown among S&P Capital IQ public firms, stratified by lagged Debt/EBITDA buckets (details in Section [IA.7.2](#)). Panel B uses macro and credit-cycle factors. Standard errors are clustered by quarter, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

Panel A: Public Firm Average Drawdown				
Debt/EBITDA:	[0, 2)	[2, 4)	≥ 4	< 0
	(1)	(2)	(3)	(4)
<i>%$\Delta Unfunded$</i>				
$\hat{\beta}$	-0.038	-0.072	-0.161	-0.072***
t	(-1.329)	(-1.573)	(-1.543)	(-5.722)
R^2	0.001	0.003	0.003	0.002
<i>$\mathbf{1}(\Delta Unfunded < 0)$</i>				
$\hat{\beta}$	0.015	0.096***	0.287***	0.066***
t	(0.387)	(2.733)	(3.107)	(4.291)
R^2	0.000	0.005	0.007	0.001
N	41,400	41,400	41,400	41,400

Panel B: Market-Wide Factors				
Factors:	ΔBBB	ΔHY	SP500	ΔVIX
	(1)	(2)	(3)	(4)
<i>%$\Delta Unfunded$</i>				
$\hat{\beta}$	-0.060**	-0.025*	0.282*	-0.003
t	(-2.263)	(-1.852)	(1.780)	(-1.518)
R^2	0.004	0.003	0.003	0.003
<i>$\mathbf{1}(\Delta Unfunded < 0)$</i>				
$\hat{\beta}$	-0.075***	-0.033***	-0.340**	-0.004**
t	(-3.667)	(-2.971)	(-2.294)	(-2.250)
R^2	0.005	0.005	0.003	0.004
N	41,400	41,400	41,400	41,400

Table 8. Variance Decomposition of Liquidity Shocks from Commitment Books

This table reports R^2 s from regressing borrower-level liquidity shocks on each of four fixed-effect specifications: time, industry-time, HQ state-time, and BDC-time (Time is year-quarter). The sample consists of BDC-borrower-quarters in continuing unfunded facilities (excluding originations and terminations) with $u_{i,\ell,t-1} > \$1$ million between 2005Q1 and 2025Q4. Industry is the borrower's industry classification; HQ state is the borrower's headquarters state (HQ country used for foreign borrowers). The outcome is either the continuous shock $\Delta u_{i,\ell,t}/u_{i,\ell,t-1}$ or the binary shock $\mathbf{1}\{\Delta u_{i,\ell,t} < 0\}$; OLS columns weight all observations equally, WLS columns use $u_{i,\ell,t-1}$ as weights.

	$\% \Delta \text{Unfunded}$		$\mathbf{1}(\Delta \text{Unfunded} < 0)$	
	OLS (1)	WLS (2)	OLS (3)	WLS (4)
Time FEs	0.009	0.014	0.012	0.014
Industry-Time FEs	0.041	0.071	0.062	0.108
HQ State-Time FEs	0.056	0.096	0.084	0.137
BDC-Time FEs	0.076	0.097	0.085	0.119
N	53,038	53,038	53,038	53,038

Table 9. Relationship Between Unfunded Commitments and Bank Credit Lines

This table regresses the amount of bank credit lines on the BDC's total unfunded commitment to borrowers in its commitment book. The sample consists of 4,695 BDC-quarters in 2005Q1–2025Q4 with a bank credit line. In columns 1–3 the dependent variable is the BDC's total bank credit line limit summed across facilities, scaled by lagged total assets; in columns 4–6 it is the undrawn bank credit limit, also scaled by lagged total assets. The regressor Total Unfunded/ TA_{t-1} sums the unfunded commitments the BDC has extended to borrowers in its commitment book, scaled by lagged total assets; *Public* is an indicator equal to one when the BDC is currently exchange-listed; lagged controls are total debt and cash and equivalents, both scaled by lagged total assets, and log lagged total assets. Columns vary in BDC and time fixed effects (Time is year-quarter), standard errors are clustered at the BDC level, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

Dependent Variable: Bank Credit Line Capacity / TA_{t-1}						
	Limit			Undrawn		
	(1)	(2)	(3)	(4)	(5)	(6)
Total Unfunded / TA_{t-1}	1.720*** (0.252)	1.034*** (0.191)	0.903*** (0.189)	1.237*** (0.196)	0.644*** (0.194)	0.531*** (0.202)
Debt / TA_{t-1}	0.604*** (0.156)	0.679*** (0.136)	0.473*** (0.163)	0.074 (0.110)	0.217* (0.120)	0.029 (0.132)
Cash / TA_{t-1}	0.794*** (0.258)	0.320 (0.199)	0.298 (0.189)	0.629*** (0.205)	0.333* (0.178)	0.314* (0.165)
log TA_{t-1}	−0.192*** (0.021)	−0.407*** (0.025)	−0.429*** (0.026)	−0.141*** (0.016)	−0.298*** (0.019)	−0.321*** (0.021)
Public	−0.173*** (0.044)	0.166*** (0.059)	0.090 (0.068)	−0.032 (0.031)	0.209*** (0.052)	0.134** (0.056)
BDC FEs	N	Y	Y	N	Y	Y
Time FEs	N	N	Y	N	N	Y
N	4,695	4,693	4,693	4,695	4,693	4,693
R^2	0.448	0.752	0.764	0.372	0.652	0.672

Table 10. BDC Liquidity Needs and Bank Credit Lines

This table decomposes the BDC's quarterly bank credit line response to investment cash demand across four facility-level margins. The sample is BDC-quarters in 2005Q1–2025Q4 with all five cash-budget levers, the four facility-level outcomes, the regressor, and the lagged controls non-missing (the same sample as Table IA.2). The dependent variables are constructed at the facility level from the as-borrower bank credit-line panel and aggregated to the BDC-quarter: column 1 reports the active drawdown on continuing facilities (the change in outstanding); column 2 reports the change in credit limit on continuing facilities (limit amendment); column 3 reports the credit limit of newly originated facilities (inception); column 4 reports the credit limit of facilities that close (termination, signed positive as a capacity loss). All four outcomes are scaled by lagged total assets. The regressor is the BDC's gross investment cash outflow scaled by lagged total assets, identical to Table IA.2. Lagged controls are undrawn bank credit line capacity, total debt, cash and equivalents, and log assets, all scaled by lagged total assets where relevant. All specifications include BDC and time fixed effects (Time is year-quarter), standard errors are clustered at the BDC level, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

Dependent Variable: Δ Bank Credit Line Capacity / TA_{t-1}				
	Drawdown (1)	Amendment (2)	Inception (3)	Termination (4)
Inv. Cash Outflow / TA_{t-1}	0.103*** (0.016)	0.037*** (0.008)	0.059** (0.025)	-0.014 (0.010)
Undrawn Bank Credit / TA_{t-1}	0.139*** (0.019)	-0.039*** (0.009)	-0.081*** (0.027)	0.064*** (0.014)
Debt / TA_{t-1}	-0.128*** (0.024)	0.019 (0.012)	-0.107*** (0.038)	-0.014 (0.016)
Cash / TA_{t-1}	0.006 (0.029)	-0.007 (0.019)	0.009 (0.050)	-0.001 (0.019)
log TA_{t-1}	-0.001 (0.004)	-0.003 (0.002)	-0.029*** (0.007)	-0.003 (0.003)
BDC FEs	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y
N	3,755	3,755	3,755	3,755
R^2	0.303	0.182	0.237	0.234

Appendix A Model Proofs

Throughout, we verify $\bar{R}_{IC}(m) \leq y - \Delta$, bank participation and BDC participation ex post to simplify the analysis. The first two are satisfied by exogenous parameters: Bank participation is satisfied if the exogenous rate R_I charged on BDC is sufficiently high. As for the BDC participation constraint, it imposes a lower bound on R_N .

In addition, we focus on the parameter region

$$y - R_I \rho > 0.$$

The inequality ensures that absent bankruptcy costs, the BDC provides credit lines.

A.1 Proof of Proposition 1

Proof. For the analysis, we treat m as a continuous variable. The objective is piecewise smooth in m , with derivative jumps at integer crossings of $\hat{k}(m)$; we analyze the FOC away from these crossings and incorporate the jumps separately. We further impose $R_N^* = \bar{R}_{IC}(m)$ and verify ex post that $R_N^* \leq y - \Delta$ and that R_N^* exceeds the lower bound implied by the BDC participation constraint.

The default threshold is defined by

$$R_N(n - \hat{k}) + \min\{m, \hat{k}\}(y - \Delta) - R_I \left[n + \min\{m, \hat{k}\} \rho \right] = 0. \quad (\text{A.1})$$

This yields two cases. If $m \geq \hat{k}$, then

$$\hat{k} = \frac{(R_I - R_N)n}{y - \Delta - R_N - R_I \rho}. \quad (\text{A.2})$$

If $m < \hat{k}$, then

$$\hat{k} = \frac{n(R_N - R_I) + m[(y - \Delta) - R_I \rho]}{R_N}. \quad (\text{A.3})$$

Case 1: $m \geq \hat{k}$. In this region, $\hat{k}$ is given by (A.2) and independent of m . The BDC's objective can be written as

$$\begin{aligned} \max_m \sum_{k \leq \hat{k}} \mathbb{P}(k) & \left\{ \bar{R}_{IC}(m)(n - k) + \min\{m, k\}(y - \Delta) - R_I[n + \rho \min\{m, k\}] \right\} \\ & + (1 + \lambda) \sum_{k > \hat{k}} \mathbb{P}(k) \left\{ \bar{R}_{IC}(m)(n - k) + \min\{m, k\}(y - \Delta) - R_I[n + \rho \min\{m, k\}] \right\}. \end{aligned} \quad (\text{A.4})$$

Differentiating with respect to m and using $\frac{\partial \min\{m, k\}}{\partial m} = \mathbf{1}_{\{k > m\}}$,

$$\begin{aligned} F(m) &= \sum_{k \leq \hat{k}} \mathbb{P}(k) (n - k) \frac{d\bar{R}_{IC}(m)}{dm} + (1 + \lambda) \sum_{\hat{k} < k \leq m} \mathbb{P}(k) (n - k) \frac{d\bar{R}_{IC}(m)}{dm} \\ &\quad + (1 + \lambda) \sum_{k > m} \mathbb{P}(k) \left\{ (n - k) \frac{d\bar{R}_{IC}(m)}{dm} + [y - \Delta - R_I \rho] \right\} \\ &= \frac{d\bar{R}_{IC}(m)}{dm} \sum_k \mathbb{P}(k)(n - k) + \lambda \frac{d\bar{R}_{IC}(m)}{dm} \sum_{k > \hat{k}} \mathbb{P}(k)(n - k) \\ &\quad + (1 + \lambda) [y - \Delta - R_I \rho] \mathbb{P}(k > m). \end{aligned} \quad (\text{A.5})$$

To simplify, note that

$$\begin{aligned} \frac{d\bar{R}_{IC}(m)}{dm} &= \frac{d}{dm} \left[(1 - q)y + qR_B + \frac{p\Delta}{1 - p} (s(m) - q) \right] \\ &= \frac{\Delta}{n(1 - p)} \frac{d}{dm} \left(\sum_{k \leq m} \mathbb{P}(k) k + \mathbb{P}(k > m) m \right) \\ &= \frac{\Delta \mathbb{P}(k > m)}{n(1 - p)}. \end{aligned} \quad (\text{A.6})$$

Substituting this together with $\sum_k \mathbb{P}(k)(n-k) = n(1-p)$, the FOC simplifies to

$$F(m) = P(k > m) \left\{ \Delta + \Delta \cdot \lambda \sum_{k > \hat{k}} P(k) \frac{1 - \frac{k}{n}}{1 - p} + (1 + \lambda)[y - \Delta - R_I \rho] \right\}. \quad (\text{A.7})$$

Hence, the sign of $F(m)$ is determined by the curly bracketed term alone for any $m \geq \hat{k}$. This term trades off the benefit of extracting more rents from firms by raising m and therefore improving coverage, and the cost of liquidity injection ex post in the distressed states. Generically the bracketed term is nonzero, so the optimum is not interior to this region: if $F(m = \hat{k}) < 0$ the objective is decreasing on $[\hat{k}, n]$ and $m^* < \hat{k}$ (moves to Case 2); if $F(m = \hat{k}) > 0$ the objective is increasing and the maximum in this region is attained at $m = n$.

Case 2: $m < \hat{k}$. In this case, $\hat{k}$ is given by (A.3) and depends on m . First, we fix $\hat{k}$ away from integer crossings, so the effects of m on $\hat{k}$ are ignored. Following the same approach as in Case 1, the first-order condition in the region $m < \hat{k}$ is

$$F(m) = \Delta P(k > m) \left[1 + \lambda \sum_{k \geq \hat{k}} P(k) \frac{1 - \frac{k}{n}}{1 - p} \right] + [y - \Delta - R_I \rho] [P(k > m) + \lambda P(k > \hat{k})]. \quad (\text{A.8})$$

The first pricing term is positive: A higher m raises $\bar{R}_{IC}(m)$ by improving coverage, and this allows the BDC to charge a higher rate. It shares the same structure as in Case 1 because the benefit of raising m to firms occurs only when $k > m$. The second is negative and captures the cost of liquidity injection ex post in states where liquidity is constrained. A wedge arises between the two cases because in Case 2, raising m also worsens the distance to default.

Now we verify that the second-order condition (SOC) holds in the smooth region away

from integer crossings $\hat{k}(m)$. Take derivative with respect to m :

$$F'(m) = \underbrace{\frac{d\mathbb{P}(k > m)}{dm}}_{-} \cdot \underbrace{\left\{ \Delta \left[1 + \lambda \sum_{k \geq \hat{k}} \mathbb{P}(k)^{\frac{1-\frac{k}{n}}{1-p}} \right] + [y - \Delta - R_I \rho] \right\}}_{\equiv A} \quad (\text{A.9})$$

Rearranging, $A = [y - R_I \rho] + \Delta \lambda \sum_{k \geq \hat{k}} \mathbb{P}(k)^{\frac{1-\frac{k}{n}}{1-p}} > 0$, because $y - R_I \rho > 0$ by assumption (under which the BDC provides credit lines absent bankruptcy frictions) and the second term is nonnegative.

We evaluate the FOC at the endpoints of the region, $m = 0$ and $m = \hat{k}^-$. At $m = 0$, $\hat{k}(0) = n$, so

$$\text{sgn}\{F(0)\} = \text{sgn}\{\Delta \mathbb{P}(k > 0) + [y - \Delta - R_I \rho] \mathbb{P}(k > 0)\} = \text{sgn}\{y - R_I \rho\} > 0, \quad (\text{A.10})$$

where the sign follows because of the assumption $y - R_I \rho > 0$, under which the BDC provides credit lines absent bankruptcy frictions. At $m = \hat{k}^-$, the FOC reduces to

$$F(\hat{k}^-) = \mathbb{P}(k > \hat{k}) \left\{ \Delta \left[1 + \lambda \sum_{k \geq \hat{k}} \mathbb{P}(k)^{\frac{1-\frac{k}{n}}{1-p}} \right] + (1 + \lambda) [y - \Delta - R_I \rho] \right\}, \quad (\text{A.11})$$

which matches $F(\hat{k})$ from Case 1 up to the positive factor $\mathbb{P}(k > \hat{k})$. This expression is negative when λ is sufficiently large, under which $m \geq \hat{k}$ are dominated. We focus on this case henceforth, because otherwise, $F(m) \leq 0$ for $m \geq \hat{k}$ and we need to discuss $m = n$ as another local maximum.

It remains to check integer crossings of $\hat{k}(m)$ since m affects default thresholds in Case 2. Let

$$m^* \equiv \inf\{m \in (0, \hat{k}) : F(m) \leq 0\}, \quad (\text{A.12})$$

i.e., the first point at which F reaches zero from above. By construction, the BDC's value is increasing for $m < m^*$. It remains to verify that integer crossings of $\hat{k}(m)$ for $m \in [m^*, \hat{k})$ imply downward jumps in the derivative. This property combined with the second-order condition implies that m^* is the unique local maximum in the interior region. At a crossing m_0 , the derivative jumps by

$$\lambda \mathbb{P}(\hat{k}) \left[\Delta \mathbb{P}(k > m) \frac{1 - \frac{\hat{k}}{n}}{1 - p} + y - \Delta - R_I \rho \right], \quad (\text{A.13})$$

Substituting $\hat{k}$ from (A.3) into $1 - \frac{\hat{k}}{n}$, the derivative jumps downward whenever the pricing gain from greater coverage is dominated by the marginal draw cost,

$$\Delta \mathbb{P}(k > m) \frac{1 - \frac{\hat{k}}{n}}{1 - p} < R_I \rho - (y - \Delta), \quad \text{for } m \in [m^*, \hat{k}). \quad (\text{A.14})$$

Note that the imposed condition does not affect key model implications and just simplifies the number of potential local maxima. It says the upper-tail probability beyond the interior optimum is small.

Taken together, if the BDC is subject to a sufficiently large bankruptcy cost, then it rations the credit lines it extends to borrower firms by choosing a small bank credit line, and $m^* < \hat{k}$. $\square$

A.2 Proof of Proposition 2

Proof. Under the Bernoulli structure $X_i = Z\xi_i$ with $Z \sim \text{Bernoulli}(z)$ and $\xi_i \sim \text{Bernoulli}(\eta_i)$, firm i is hit with probability $p_i = z\eta_i$. Conditional on being hit ($Z = 1$), its coverage probability is

$$s_i(m; \eta_{-i}) = \sum_{r=0}^{n-1} \min\left\{1, \frac{m}{1+r}\right\} \mathbb{P}(K_{-i} = r \mid Z = 1), \quad (\text{A.15})$$

where K_{-i} is the number of other firms hit.

Individual FOC. Firm i solves

$$\max_{p_i} (1 - p_i)(y - R_N) + p_i s_i(m; p_{-i}) \Delta - c(1 - p_i), \quad (\text{A.16})$$

taking p_{-i} as given. Parametrizing $p_i = z\eta_i$ and using $\partial/\partial\eta_i[z\eta_i s_i(m; \eta_{-i})] = z s_i(m; \eta_{-i})$ (since s_i does not depend on η_i), the FOC reduces to

$$c'(1 - p_i) = (y - R_N) - \Delta s_i(m; \eta_{-i}). \quad (\text{A.17})$$

Imposing symmetry $\eta_i = \eta$, $s_i(m; \eta_{-i}) = s(m)$, so $c'(1 - p^{eq}) = (y - R_N) - \Delta s(m)$.

Planner's FOC. A planner maximizing $\sum_i [(1 - p_i)(y - R_N) + p_i s_i(m; p_{-i}) \Delta - c(1 - p_i)]$ internalizes how η_i affects coverage for other firms $j \neq i$. The FOC is

$$c'(1 - p_i) = (y - R_N) - \Delta s_i(m; \eta_{-i}) - \Delta \sum_{j \neq i} \eta_j \frac{\partial s_j(m; \eta_{-j})}{\partial \eta_i}. \quad (\text{A.18})$$

We sign $\partial s_j / \partial \eta_i$ via a first-order stochastic dominance argument. Let $g(r) \equiv \min\{1, m/(1 + r)\}$, which is non-increasing in r . Conditional on $Z = 1$, decompose $K_{-j} = \xi_i + K_{-i, -j}$ where $K_{-i, -j} \equiv \sum_{\ell \neq i, j} \xi_\ell$ is independent of ξ_i . Then

$$s_j(m; \eta_{-j}) = \mathbb{E}[g(K_{-j}) \mid Z = 1] = \eta_i \mathbb{E}[g(1 + K_{-i, -j})] + (1 - \eta_i) \mathbb{E}[g(K_{-i, -j})], \quad (\text{A.19})$$

so

$$\frac{\partial s_j}{\partial \eta_i} = \mathbb{E}[g(1 + K_{-i, -j}) - g(K_{-i, -j})] \leq 0, \quad (\text{A.20})$$

with each realization $g(1 + r) - g(r) \leq 0$ since g is non-increasing. The inequality is strict iff $g(1 + r) < g(r)$ with positive probability under $K_{-i, -j}$, i.e., iff $m < r + 2$ for some

$r \in \{0, \dots, n-2\}$ in the support. Under interior $\eta_\ell \in (0, 1)$, the support is $\{0, \dots, n-2\}$, so this is equivalent to $m < n$. When $m = n$, $g \equiv 1$ on $\{0, \dots, n-1\}$ and $\partial s_j / \partial \eta_i = 0$.

In particular, when $\lambda > \hat{\lambda}$ and $m^* < \hat{k} \leq n$ by Proposition 1, the externality is strictly negative; when $\lambda < \hat{\lambda}$ and $m^* = n$, it vanishes.

Comparison. Under symmetry, the planner's FOC becomes

$$c'(1 - p^{planner}) = (y - R_N) - \underbrace{\Delta s(m) - \Delta(n-1)\eta \frac{\partial s_j}{\partial \eta_i} \Big|_{\text{sym}}}_{\geq 0}. \quad (\text{A.21})$$

Since c' is strictly increasing (convex c), comparing (A.17) at $\eta = \eta^{eq}$ with (A.21) at $\eta = \eta^{planner}$:

- If $s(m) < 1$ (equivalently $m < \hat{k}$ and $\mathbb{P}(k > m) > 0$, which holds under $\lambda > \hat{\lambda}$ by Proposition 1), the externality term is strictly positive, so $c'(1 - p^{planner}) > c'(1 - p^{eq})$ and hence $1 - p^{planner} > 1 - p^{eq}$, i.e., $p^{planner} < p^{eq}$.
- If $s(m) = 1$ (i.e., $\lambda < \hat{\lambda}$ and $m^* = n$), coverage never binds, $\partial s_j / \partial \eta_i = 0$, and the two FOCs coincide.

□

Internet Appendix

“Credit Commitments by Nonbanks”

IA.6. Data Collection and Processing

This section provides details on sample construction. Section [IA.6.1](#) defines the universe of BDCs and their regulatory filings. Sections [IA.6.2–IA.6.5](#) describe each group of variables collected from those filings: total unfunded commitments, borrower-level commitment books, bank credit lines, balance-sheet items, and statement-of-cash-flow items. Section [IA.6.6](#) describes the linking of borrowers in BDC commitment books to PitchBook companies. Section [IA.6.7](#) describes the FFIEC Call Report data on banks.

IA.6.1. Universe of BDCs from SEC EDGAR

We identify the universe of business development companies from the SEC’s EDGAR system based on Form N-54A, N-54B, and N-54C filings, which respectively elect, amend, and withdraw BDC status under the Investment Company Act of 1940. The BDC universe over our sample period (2005Q1–2025Q4) consists of all entities with an active N-54A election in any given quarter. The SEC’s annual public BDC list is used to discover CIKs but is incomplete as a status authority, so we additionally filter on N-54A/N-54C dates: 225 CIK-quarter filings from 11 entities are removed because they fall after a recorded N-54C withdrawal (so the entity had ceased to be a BDC at the time of the filing), and 12 “ghost” CIKs that appear on SEC lists with no corresponding 10-K/10-Q filings are dropped as data-entry artifacts. The cleaned list of filings contains 6,747 CIK-quarter filings across 245 unique BDCs spanning 2005Q1–2025Q4. For each CIK-quarter in this roster we collect the 10-K or 10-Q filing from EDGAR’s full-text search index; these filings are the source for the

variables described in Sections [IA.6.2–IA.6.5](#) below.

Within this universe, we further distinguish BDCs by exchange-listing status: Listed BDCs (file Form 8-A12B and trade on exchanges) and non-listed public BDCs (file periodic reports under Section 15(d) or 12(g) but are not listed on a national exchange). Listing status is assigned by matching each filing to the most recent Form 8-A registration or Form 15 deregistration event on the CIK.¹ Figure [IA.4](#) shows the growth of the two public BDC segments by count and aggregate total assets. The exchange-listed segment grew from 2 BDCs in 2005Q1 to roughly 50 by the mid-2010s and has been stable since, with aggregate total assets rising from near zero to about \$180 billion by 2025Q4. The non-listed segment grew modestly through the 2010s but expanded sharply from 2020Q1 onward, reaching about 100 BDCs and roughly \$370 billion in aggregate total assets by 2025Q4, which reflects the recent proliferation of perpetual non-traded BDCs raising capital through registered public offerings. The two segments combined now manage roughly \$550 billion of total assets.

IA.6.2. BDC-Level Total Unfunded Commitments

During our sample period, nearly all BDCs disclose in their 10-K/10-Q the aggregate dollar value of unfunded loan commitments to portfolio companies, typically in the commitments-and-contingencies footnote or the management discussion section. We manually collect this filing-level total amount for each CIK-quarter. In the process, we ensure it is not confused with undrawn credit limits of bank credit lines in which the BDC is a borrower.

IA.6.3. Unfunded Credit in Commitment Books

A subset of BDC-quarters disclose a table or list of the BDC’s unfunded commitments to each portfolio company, typically in the same commitments-and-contingencies note as the

¹Private BDCs that elect 1940 Act BDC status without registering under the Exchange Act file no periodic reports with the SEC and are therefore outside the scope of our sample.

total amount. We collect each borrower’s unfunded amount and commitment type (e.g., revolving credit facility, delayed-draw term loan), along with the corresponding outstanding loan principal where reported. For quality control, we validate the sum of unfunded amounts within a book against the BDC’s reported total unfunded commitments (Section [IA.6.2](#)). The dataset we collect covers 2,989 BDC-quarters and 138,456 facility-level observations within our 2005Q1–2025Q4 sample window.

IA.6.4. BDCs’ Bank Credit Lines

BDCs’ 10-K and 10-Q filings discuss each bank credit facility the BDC has as a borrower in a dedicated borrowings note. The note typically gives, for each active facility at the filing date, the facility name, the lender name (the administrative agent and other lenders if syndicated), the credit limit, the drawn balance, the undrawn capacity, the maturity date, the interest-rate terms (benchmark and spread), and any contractually specified accordion provision (the maximum permitted limit increase). Discrete changes to a facility during the quarter—origination of a new facility, amendment to an existing facility, exercise of an accordion, termination, or replacement—are reported in the same note, generally as narrative text linked to the facility name.

We assign a unique credit-line identifier (CLID) to each bank credit facility a BDC discloses. The CLID identifies a facility across time, even if the facility experiences changes at different filing dates across quarters. Sub-facilities of an SPV credit structure that are disclosed under separate names with separate limits are each given their own CLID. Renamings of the same facility (e.g., after a replacement of the administrative agent) are kept on the same CLID with an event flag for the rename. For each CLID-quarter pair, we record the values of the variables observed in the filing at that date: credit limit, drawn balance, undrawn capacity, maturity date, administrative agent, other syndicate lenders, interest-rate bench-

mark, spread, and any accordion provision. The resulting panel covers 9,055 CLID-quarter rows across 756 unique CLIDs and 198 BDCs spanning 2005Q1–2025Q4.

We record events from two complementary sources. The inceptions and terminations of CLIDs are derived directly from the CLID-quarter panel: the inception of a CLID is the first quarter in which that CLID appears, and the termination is the last quarter in which the CLID appears (excluding the sample end). Amendments and accordion exercises, which change the value of a variable on an existing CLID without changing CLID identity, are extracted from the same borrowings note as the variables: each disclosed change is recorded as a separate event with the event type (amendment, accordion exercise, replacement of admin agent, etc.), the changed field (limit, maturity, spread, etc.), and the pre- and post-change values.

IA.6.5. Balance Sheet and Cash Flow Statement Items

We collect each BDC’s balance-sheet items, including total assets, total liabilities, total equity, total debt, and cash and equivalents, from the 10-K/10-Q filings. We supplement with Compustat and S&P Capital IQ data on the same CIK and quarter and use them to cross-validate against the filing-derived values.

We also collect statement-of-cash-flow (StoCF) data from the filings. Seven items are extracted: the three category subtotals (net operating, investing, and financing cash flows) and four line items (gross investment purchases, gross investment sales and repayments, distributions paid, and equity issuance). This cash flow dataset covers 6,294 BDC-quarter rows across 242 BDCs spanning 2005Q1–2025Q4.

IA.6.6. Identifying Borrowers in Unfunded Commitment Books

The unfunded commitment book data in Section [IA.6.3](#) include borrower *names*. However, without a unique identifier, we cannot (i) track each borrower across quarters and BDCs and align the unfunded commitment book with the BDC’s funded holdings, and (ii) merge in firm characteristics (industry, state, age, sponsor backing) for the analyses. We use PitchBook’s company identifier for this. To do so, a challenge is that PitchBook tracks borrowers by parent/operating company, while filings disclose the specific borrowing entity—often an SPV, intermediate holdco, or acquisition shell (“ABC Buyer, Inc.” rather than “ABC Corp”). Direct name matching fails on a non-trivial share of borrowers even when the underlying firm is in PitchBook.

Our linking strategy uses FactSet’s BDC portfolio holdings data as a bridge. FactSet records each holding under the original borrowing-entity name from the EDGAR filing, so the EDGAR name and the FactSet name are near-identical. FactSet also reports the instrument-level terms (par value, maturity, base rate, spread) for each holding, and PitchBook reports the same instrument-level terms under its own parent-keyed company identifier. For each holding we match the FactSet record to a PitchBook record on a composite key of (par value, maturity, base rate, spread), which works precisely because it identifies the same loan from the SPV side (FactSet) and the parent side (PitchBook). The PitchBook identifier from the matched holding is then carried back onto the corresponding borrower name in the commitment book. For the remaining unmatched borrowers, we apply a within-CIK name matching to PitchBook. The combined procedure identifies 84–87% of unique borrower names.

IA.6.7. Bank Data from FFIEC Call Reports

We obtain data on banks from the FFIEC’s Consolidated Reports of Condition and Income (Call Reports). We extract balance-sheet items (Schedule RC), loans and leases by category (RC-C Part I), and off-balance-sheet unused commitments (RC-L) for each bank-quarter. The unused-commitment items require one reconciliation: in December 2024 the FFIEC retired the combined financial-institution unused-commitment line (RCFDJ458) and replaced it with separate depository (PV10) and nondepository (PV11) components starting 2024Q4. For a time-stable measure we use RCFDJ458 prior to the revision and $PV10 + PV11$ thereafter, and we expose PV11 separately for analyses that need to isolate nondepository (BDC-relevant) from interbank commitments.

IA.7. Measurement of Credit Facility Drawdown

IA.7.1. Borrower Drawdown in BDC Unfunded Commitment Books

Our sample of BDC unfunded commitment books consists of continuing facilities, defined as positions present in both $t - 1$ and t in the same combination of BDC-borrower-commitment type. This filter excludes originations and terminations of credit facilities, which would otherwise mechanically inflate $\% \Delta u$ or generate $\% \Delta u = -1$. We additionally restrict the sample to BDC-borrower-quarters with lagged unfunded commitment above \$1 million. The restriction avoids small-denominator noise in $\% \Delta u$. The threshold matches the analogous filter applied to S&P Capital IQ public-firm drawdowns in Section [IA.7.2](#).

We measure the borrower’s liquidity shock with two alternative measures. The continuous shock is the percentage change in unfunded commitment, $\% \Delta u_{i,\ell,t} = \Delta u_{i,\ell,t} / u_{i,\ell,t-1}$, winsorized at the 1st and 99th percentiles. The binary shock is the indicator $\mathbf{1}\{\Delta u_{i,\ell,t} < 0\}$. The continuous measure helps quantify the magnitude of drawdown, while the binary mea-

sure is robust to the outlier tail that the continuous measure inherits and is more interpretable as the fraction of borrowers actively drawing down their commitment.

The interpretation of Δu requires care because changes in unfunded commitment can come from two sources: (i) borrower drawdowns and repayments; (ii) amendments to the total committed amount. Because the BDC’s total commitment to a borrower is not consistently reported in 10Q and 10K filings, we cannot always separately identify the two sources of change in unfunded commitment. However, in a subsample where the total commitment is reported (15 BDCs, 5,212 BDC-borrower-quarters), we find that source (ii) is infrequent: amendments to the total commitment occur in only 21.7% of BDC-borrower-quarter observations, suggesting that Δu is a reasonable proxy for borrower drawdown activity. Figure [IA.5](#) shows that when the total commitment is unchanged between two consecutive quarters (78.3% of observations), the accounting identity $\Delta u = -\Delta \text{funded}$ holds exactly, with a sample correlation of -1.000 .

IA.7.2. Public-Firm Drawdown Factors from S&P Capital IQ

Regressions in Table [7](#) use aggregate drawdown factors constructed from S&P Capital IQ (CIQ) Capital Structure data merged with the Compustat quarterly database. The sample is non-financial Compustat firms (SIC outside 6000–6999) over 2010Q1–2024Q4, restricted to firm-quarters with lagged credit-line drawn balance above \$1 million. This yields approximately 79,000 firm-quarters across around 4,600 unique firms. For each firm-quarter, we measure drawdown as the percentage change in outstanding credit line drawn amount, winsorized at the 1st and 99th percentiles. We use change in drawn rather than change in unfunded because CIQ does not consistently report credit limits, particularly in fourth-fiscal quarters. We then stratify firms by their lagged Debt-to-EBITDA ratio (i.e., the sum of long-term and current-portion long-term debt, divided by trailing-four-quarter EBITDA)

into four buckets ($[0, 2)$, $[2, 4)$, ≥ 4 , and a negative-EBITDA bucket). For each bucket and quarter, we compute the dollar-weighted average drawdown (weighting by lagged drawn balance). The resulting four quarterly time series are the factors used in columns 1–4 of Table 7, Panel A.

IA.8. Bank Credit Lines: Composition, Events, and Robustness

This appendix documents the bank credit line data underlying Section 3.3 and reports a robustness test of the backstop interpretation. Our data on credit-agreement details (i.e., lenders, credit limits, drawn balances, maturities, spreads, and accordion provisions) cover a panel of 9,055 CLID-quarter observations across 756 credit lines and 198 BDCs.

Table IA.5 summarizes the lenders that provide these credit lines. After standardizing lender names and merging bank-holding-company broker-dealers into their parent type, US and foreign banks together provide the vast majority of credit lines to BDCs. The administrative-agent role is concentrated among a small set of large banks, led by JPMorgan, Deutsche Bank, and Citibank. Figure IA.6 plots the composition of administrative agents by lender type at annual frequency over 2005–2025, showing that the predominance of bank lenders has persisted throughout the sample period.

Table IA.6 reports the contractual events on these facilities: of 6,584 events, amendments are the most common, and amendments overwhelmingly upsize credit limits and extend maturities—facilities ratchet up as BDCs scale. Table IA.7 documents the accordion feature: about 30% of credit lines carry an accordion option that allows the BDC to upsize the facility without negotiating a new agreement, roughly half of those options are exercised at least once, and the median accordion cap sits about 30% above the base credit limit.

Finally, Table IA.3 reports a panel event study around the 2023 regional bank stress. We test whether BDCs with greater pre-shock exposure to US regional banks, or to admin-

istrative agents with low Tier-1 capital ratios, experience contraction in their bank credit line limits after 2022Q4. Across both treatment definitions and all time horizons, the estimated effects are statistically insignificant with flat pre-trends. This result shows that pre-committed revolving facilities remain in place even when the lending bank itself is under stress, supporting the liquidity backstop interpretation.

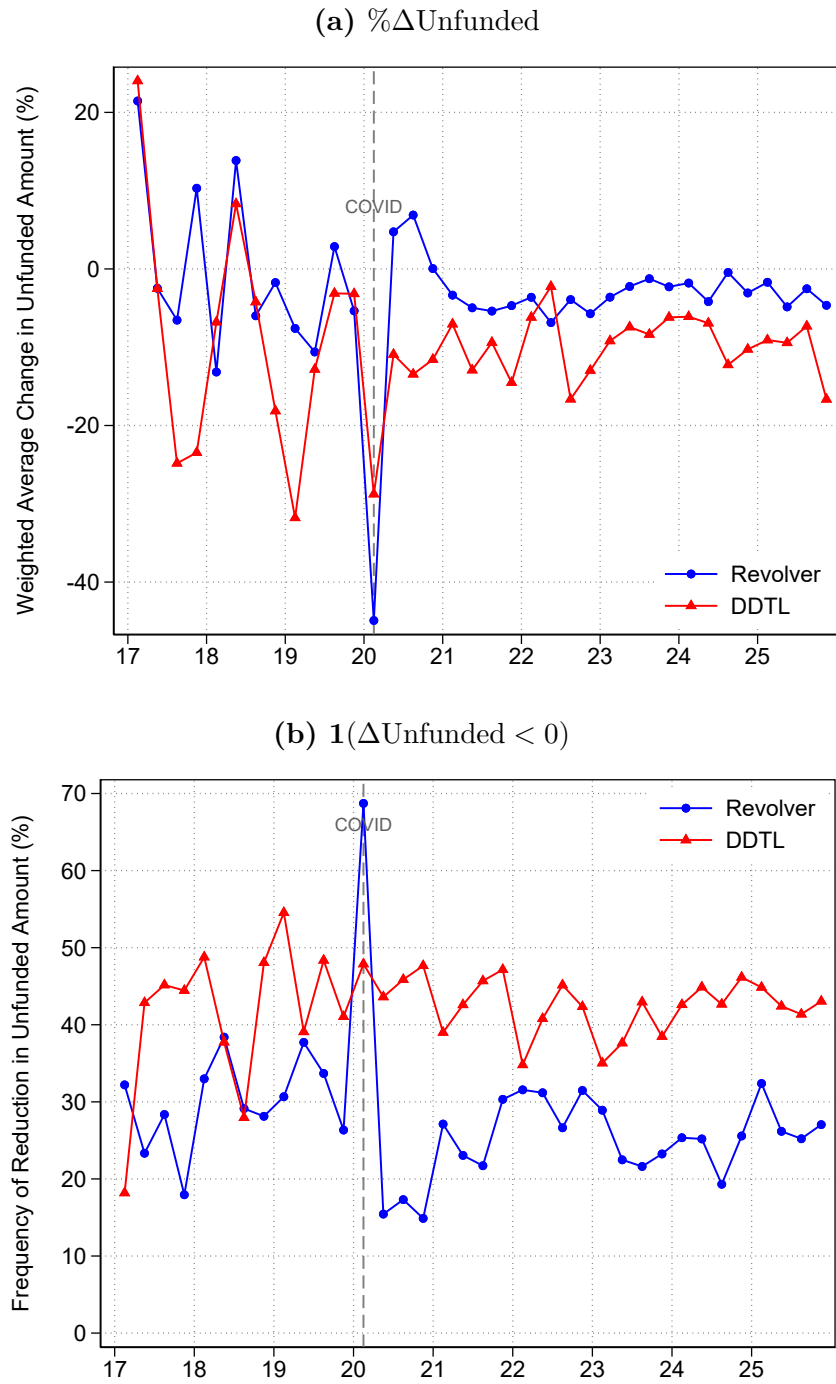


Figure IA.1. Changes in Unfunded Commitments by Commitment Type.

This figure presents the time series of changes in unfunded commitments extended by BDCs to their borrowers from 2017Q1 to 2025Q4, categorized by commitment type (Revolver and DDTL). Panel (a) shows the average quarterly percentage change in unfunded commitment amounts for each type. Panel (b) displays the quarterly frequency of reductions in unfunded commitment amounts. The vertical dashed line marks the onset of the COVID-19 pandemic.

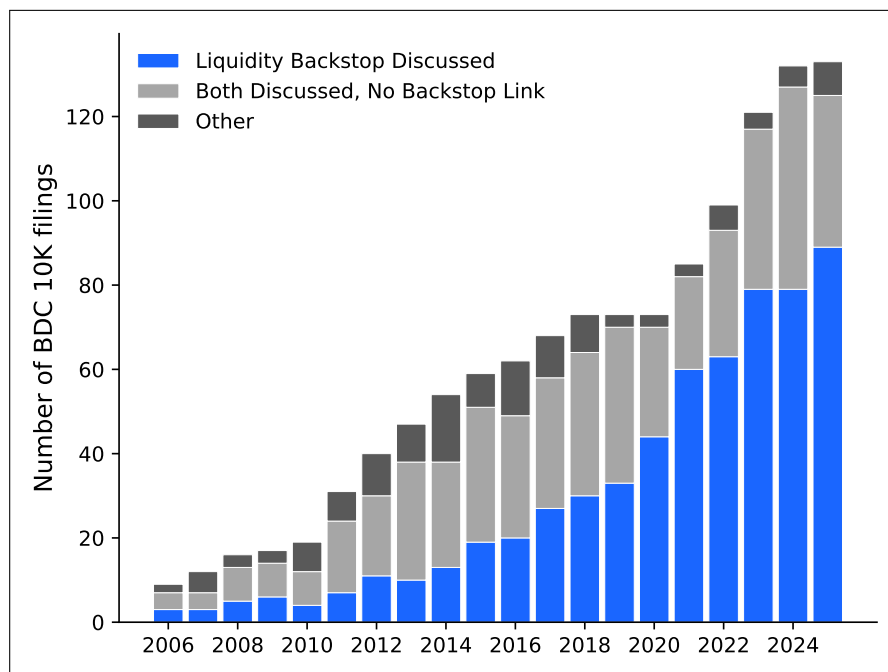


Figure IA.2. Disclosure of Liquidity-Backstop Discussion in BDC 10-K Filings.

This figure plots the number of BDC 10-K filings in each year by disclosure status. The sample comprises 1,243 10-K filings, each classified into one of three categories. “Liquidity Backstop Discussed” marks filings in which the BDC names at least one specific source of liquidity (e.g., cash, bank credit facility, marketable securities, LP capital commitments) as backstop for its unfunded commitments to portfolio companies. “Both Discussed, No Backstop Link” marks filings that discuss both the BDC’s unfunded commitments and its liquidity sources without explicitly connecting the two, including filings that mention only generic-adequacy language (e.g., “adequate financial resources”) without naming a specific source. “Other” includes filings with no relevant disclosure or where only one side (liquidity or commitments) is discussed.

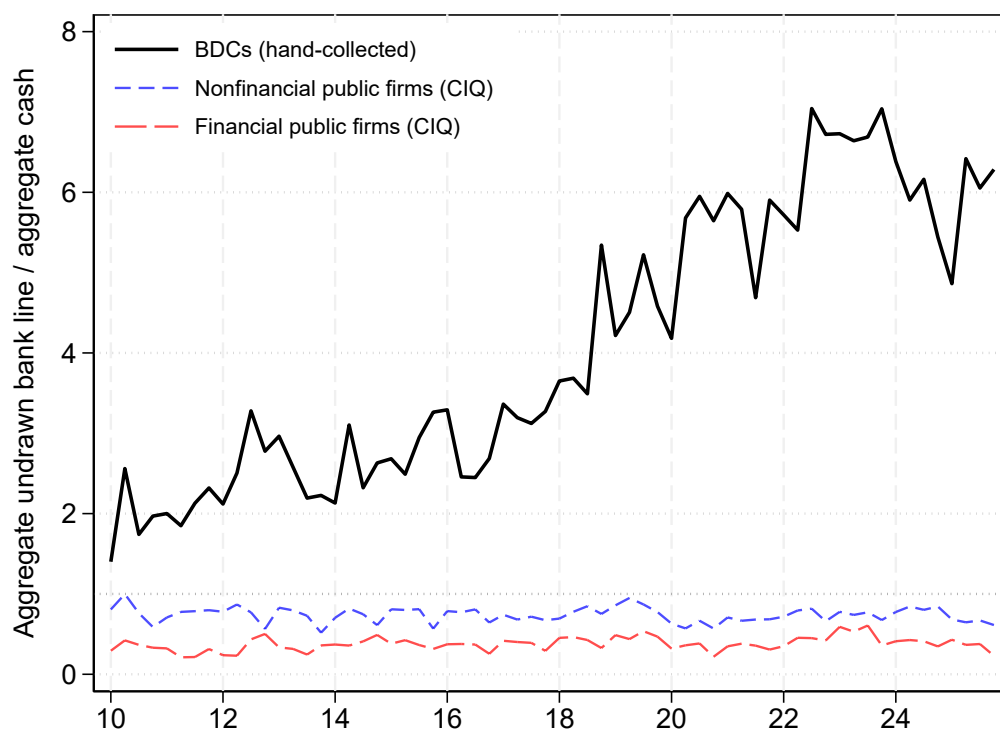


Figure IA.3. Undrawn Bank Credit to Cash Ratio: BDCs vs. Public Firms.

This figure plots the quarterly aggregate ratio of undrawn bank credit line capacity to cash and cash equivalents, separately for the BDC sector (hand-collected from EDGAR), nonfinancial public firms (Compustat S&P Capital IQ, SIC outside 6000–6999), and financial public firms (SIC 6000–6999). Within each sector and quarter, the numerator is the sum of undrawn capacity (credit limit – outstanding) across all firms that report a credit-line limit, and the denominator is the sum of cash and cash equivalents across the same firms. Firm-quarters with reported undrawn capacity exceeding total assets are dropped as data errors. The dotted horizontal line at one separates regimes in which undrawn capacity exceeds cash from those in which cash exceeds undrawn capacity. The median quarterly ratio is 3.43 for the BDC sector, 0.76 for nonfinancial public firms, and 0.37 for financial public firms. BDCs therefore hold roughly four times the undrawn-to-cash ratio of nonfinancial public firms and nine times that of financial public firms over the same period, indicating that bank credit lines are a much larger component of the BDC sector’s immediate liquidity buffer than at public firms. Data sources: SEC’s EDGAR (BDCs); S&P Capital IQ Capital Structure and Compustat quarterly (public firms).

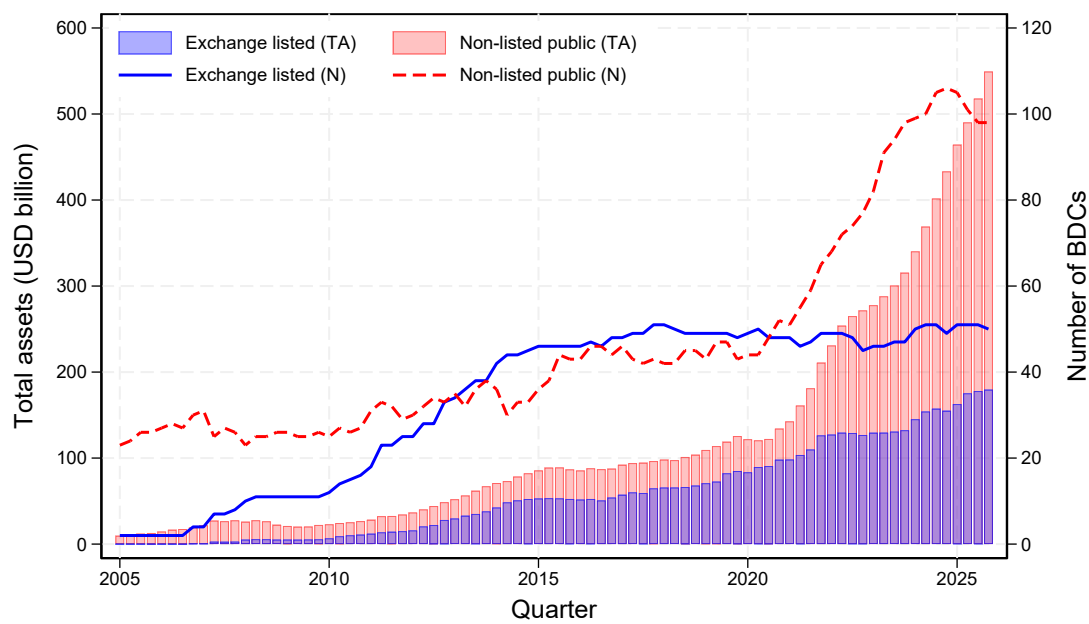


Figure IA.4. BDCs by Listing Status: Number and Total Assets.

This figure plots the BDC universe between 2005Q1 and 2025Q4 by listing status. The stacked bars (left axis) show total assets (USD billion) summed across BDCs in each segment per quarter, with exchange-listed BDCs at the bottom (blue) and non-listed public BDCs stacked on top (red). The solid blue and dashed red lines (right axis) show the corresponding number of BDCs in each segment. Data source: SEC EDGAR.

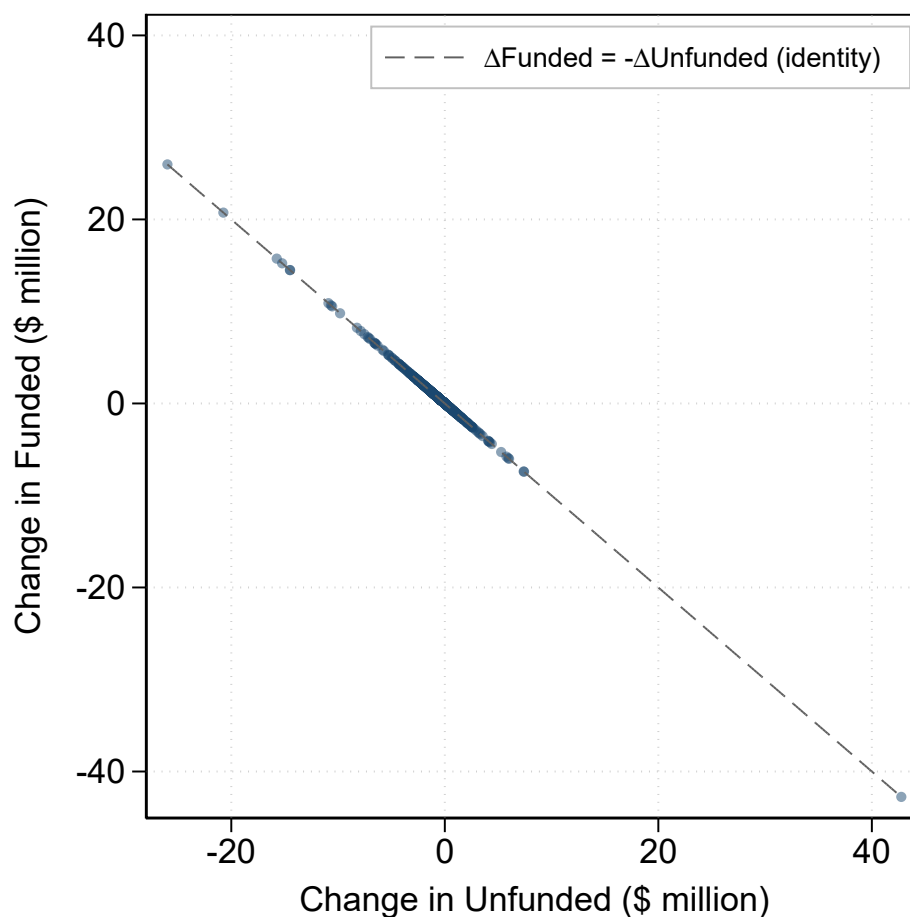


Figure IA.5. Change in Unfunded Commitment as a Proxy for Drawdown.

This figure plots quarterly changes in unfunded commitments against changes in funded commitments at the BDC–portfolio company level, within the subsample where the BDC’s total commitment to the borrower is unchanged between consecutive quarters (78.3% of observations). On this subsample, the accounting identity $\Delta\text{Funded} = -\Delta\text{Unfunded}$ holds by construction, and the points lie on the identity line with a sample correlation of -1.000 . The sample is restricted to 15 BDCs (5,212 BDC-borrower-quarters) that report funded amounts for individual portfolio company commitments.

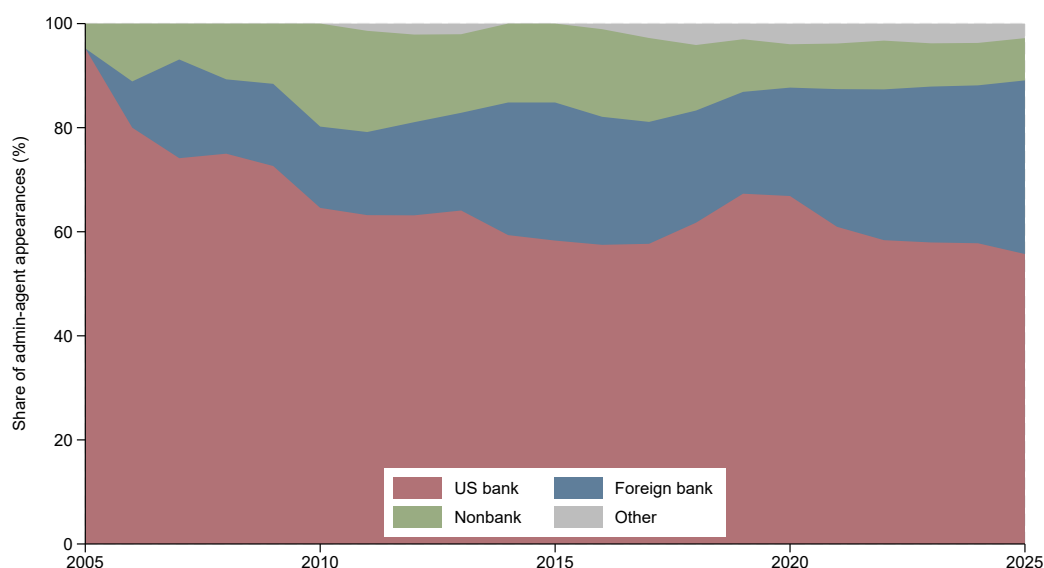


Figure IA.6. Admin-Agent Composition Over Time.

This figure plots the share of admin-agent appearances on BDC credit lines by lender type at annual frequency over 2005–2025, as a stacked area chart. The four buckets match Table IA.5: US bank, Foreign bank (foreign-headquartered banks operating in the US through a domestic branch), Nonbank (independent nonbank lenders, asset managers, and insurance companies; SEC-registered broker-dealers are reclassified into the type of their parent), and Other (BDC investment-adviser affiliates, trust companies, and central-bank discount-window borrowings). Source: SEC’s EDGAR (10-K and 10-Q filings).

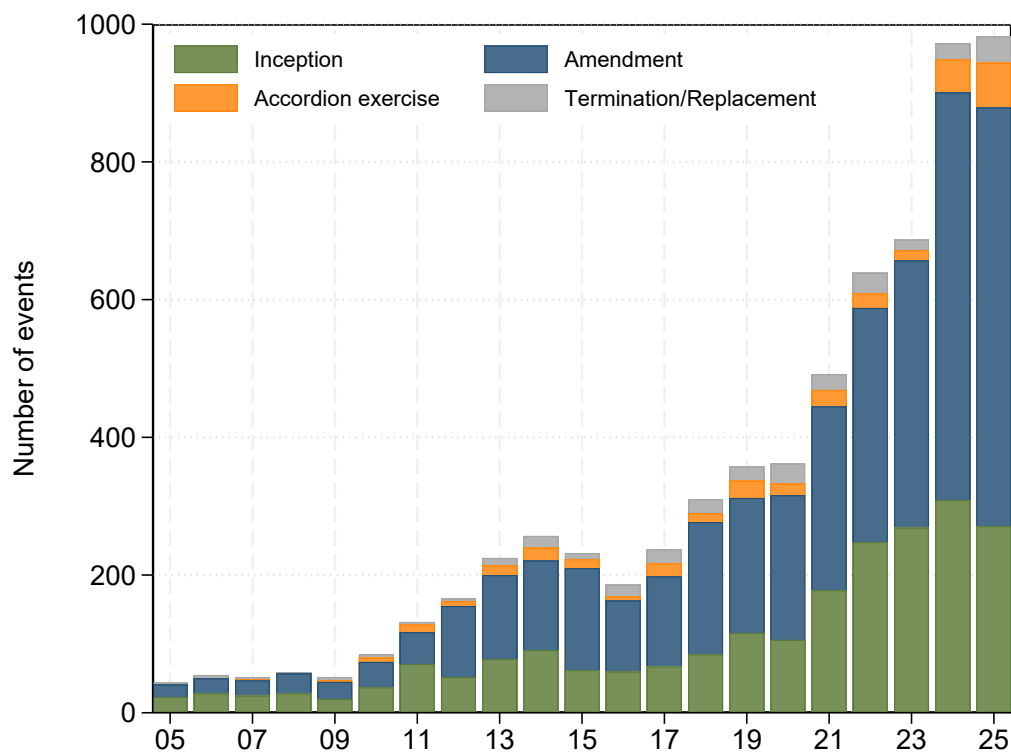


Figure IA.7. Bank Credit Line Events Over Time.

This figure plots the annual count of bank credit line events extracted from BDC 10-K and 10-Q filings, 2005–2025, stacked by event type: inception, amendment, accordion exercise, and termination/replacement (terminations and replacements are pooled). Source: SEC’s EDGAR (10-K and 10-Q filings).

Table IA.1. Summary Statistics for Liquidity-Response Regressions

This table reports summary statistics for the variables used in Table 10 and Table IA.2. The sample is the common BDC-quarter estimation sample after absorbing BDC and time fixed effects. Variables scaled by assets are divided by lagged total assets. All scaled variables are winsorized at the 1st and 99th percentiles within the pre-FE common sample, matching the regression specifications; log assets are not winsorized.

Variable	<i>N</i>	Mean	SD	P25	Median	P75
<i>Panel A. Investment demand and bank credit line response</i>						
Inv. cash outflow / TA_{t-1}	3,755	0.205	0.244	0.067	0.128	0.239
Bank credit line drawdown / TA_{t-1}	3,755	0.025	0.099	-0.010	0.000	0.046
Bank credit line amendment / TA_{t-1}	3,755	0.013	0.060	0.000	0.000	0.000
Bank credit line inception / TA_{t-1}	3,755	0.035	0.132	0.000	0.000	0.000
Bank credit line termination / TA_{t-1}	3,755	0.014	0.065	0.000	0.000	0.000
<i>Panel B. Other funding sources</i>						
Inv. proceeds / TA_{t-1}	3,755	0.092	0.146	0.034	0.068	0.113
Equity issue / TA_{t-1}	3,755	0.060	0.154	0.000	0.001	0.063
Dist. cut / TA_{t-1}	3,755	0.013	0.017	0.007	0.012	0.016
Cash use / TA_{t-1}	3,755	-0.004	0.069	-0.015	-0.001	0.013
<i>Panel C. Controls</i>						
Undrawn bank credit line / TA_{t-1}	3,755	0.196	0.209	0.053	0.151	0.266
Debt / TA_{t-1}	3,755	0.377	0.170	0.309	0.419	0.491
Cash / TA_{t-1}	3,755	0.063	0.091	0.015	0.029	0.068
log TA_{t-1}	3,755	6.533	1.539	5.739	6.581	7.503

Table IA.2. BDC Liquidity Needs and Other Sources of Liquidity

This table decomposes BDC investment cash demand across the four non-bank-credit-line funding levers, regressing each lever on the same investment-demand regressor and controls. The CL draw-down response is reported separately in Table 10 as part of the bank credit line decomposition. The sample is the same BDC-quarters in 2005Q1–2025Q4 as Table 10. Each column reports a separate regression for one funding lever, signed so positive values are sources of funds: investment proceeds from sales and principal repayments, equity issuance, distribution cuts, and balance-sheet cash use; all four levers are scaled by lagged total assets. The regressor is the BDC’s gross investment cash outflow scaled by lagged total assets; lagged controls are undrawn bank credit line capacity, total debt, cash and equivalents, and log assets, ordered by economic importance and all scaled by lagged total assets where relevant. All specifications include BDC and time fixed effects (Time is year-quarter), standard errors are clustered at the BDC level, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

Dependent Variable: Funding Amount / TA_{t-1}				
	Inv. proceeds (1)	Equity issue (2)	Dist. cut (3)	Cash use (4)
Inv. cash outflow / TA_{t-1}	0.290*** (0.036)	0.270*** (0.030)	0.003 (0.004)	−0.037*** (0.013)
Undrawn bank credit line / TA_{t-1}	−0.094*** (0.020)	−0.074*** (0.019)	−0.006*** (0.002)	0.057*** (0.013)
Debt / TA_{t-1}	−0.010 (0.032)	0.125*** (0.036)	−0.016*** (0.005)	−0.001 (0.022)
Cash / TA_{t-1}	−0.136*** (0.042)	0.060 (0.060)	−0.006 (0.005)	0.287*** (0.038)
log TA_{t-1}	−0.016** (0.008)	−0.079*** (0.007)	−0.004*** (0.001)	0.029*** (0.005)
BDC FEs	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y
N	3,755	3,755	3,755	3,755
R^2	0.441	0.633	0.328	0.207

Table IA.3. BDCs' Bank Credit Limit Around the 2023 Banking Crisis

This table reports a panel event-study test of bank credit line backstop stickiness around the 2023 banking stress. The sample is a BDC $\times$ quarter panel of 90 BDCs with at least one identified bank credit-line administrative agent at 2022Q4, observed at horizons $t = -4, \dots, +4$ relative to 2022Q4. The dependent variable is $\log(\text{Credit Limit}_{\ell,t})$. Columns 1–2 use *RegionalExposure* as the continuous treatment, defined as the BDC's pre-shock dollar credit limit at US-regional-bank admin agents scaled by 2022Q4 total assets; columns 3–4 use *LowTier1Exposure*, the analogous limit-weighted exposure to admin agents whose Tier-1 RBC ratio at 2022Q4 falls in the bottom quartile of the admin-bank universe (cutoff 10.64%, from FFIEC Call Report Schedule RC-R). Columns 2 and 4 add horizon-interacted balance-sheet controls used in Tables 9 and 10: pre-shock $\log(\text{TA})$, Debt/TA, and Cash/TA. All specifications include BDC and time fixed effects (Time = year-quarter) with the reference horizon at $t = 0$, standard errors are clustered at the BDC level, and *, **, *** represent 10%, 5%, and 1% levels of statistical significance.

Dependent Variable: $\log(\text{Credit Limit})$				
	$\text{RegionalExposure} \times \mathbf{1}\{t = j\}$		$\text{LowTier1Exposure} \times \mathbf{1}\{t = j\}$	
Horizon t	(1)	(2)	(3)	(4)
$t = -4$	-0.491 (0.391)	-0.456 (0.393)	-0.450 (0.371)	-0.447 (0.362)
$t = -3$	-0.181 (0.269)	-0.278 (0.249)	-0.174 (0.257)	-0.214 (0.248)
$t = -2$	-0.213 (0.185)	-0.233 (0.183)	-0.310 (0.193)	-0.290 (0.193)
$t = -1$	0.082 (0.072)	0.046 (0.050)	0.001 (0.047)	0.010 (0.038)
$t = +1$	0.006 (0.106)	-0.035 (0.097)	-0.098 (0.067)	-0.089 (0.068)
$t = +2$	-0.060 (0.115)	-0.097 (0.104)	-0.143* (0.079)	-0.128 (0.078)
$t = +3$	-0.028 (0.121)	-0.079 (0.121)	-0.025 (0.107)	-0.022 (0.107)
$t = +4$	0.088 (0.145)	0.069 (0.140)	0.101 (0.128)	0.117 (0.116)
BDC FEs	Y	Y	Y	Y
Time FEs	Y	Y	Y	Y
Controls $\times t$	N	Y	N	Y
N	765	765	765	765
Within R^2	0.015	0.075	0.012	0.074

Table IA.4. Illustrative Case: Cash-Flow Items at Ares Capital Around the 2020Q1 COVID Stress

This table illustrates the cash-flow items used in the regressions of Tables 10 and IA.2, taking Ares Capital—the BDC with the largest single-quarter unfunded drawdown in our sample—as a concrete example. $\Delta Unfunded$ is the quarterly change in Ares’s aggregate unfunded commitment to borrowers in its commitment book. *Inv. Cash Outflow* is the BDC’s gross investment cash outflow, the regressor in Tables 10 and IA.2. $\Delta Bank Outstanding$ is the change in bank credit line drawn balance (Table 10 column 1). $\Delta Credit Limit$ is the net change in committed bank credit line capacity from amendments, new originations, and terminations (Table 10 columns 2–4 combined). *Inv. Proceeds*, *Equity Iss.*, *Distrib. Paid*, and $\Delta Cash$ are the cash-flow-statement items that enter Table IA.2 as four of the five funding levers. All amounts in USD million.

Quarter	ΔUnf	Inv. Outflow	$\Delta Bank Out$	$\Delta Limit$	Inv. Proc.	Eq. Iss.	Distrib.	$\Delta Cash$
2019Q3	+37	2,095	+317	—	1,164	0	170	—
2019Q4	—	1,495	+969	+340	1,045	+64	174	−86
2020Q1	−839	1,781	+464	+150	1,031	−100	172	+293
2020Q2	−41	938	−786	+637	+1,633	0	169	−182
2020Q3	−89	649	−992	—	432	0	169	−45
2020Q4	+81	3,391	+936	—	2,372	+4	169	+21

Table IA.5. Lender Types of BDCs' Credit Lines

This table summarizes the lenders that provide credit lines to BDCs between 2005Q1 and 2025Q4. Lender names are extracted from 10-K and 10-Q credit-agreement disclosures, canonicalized to a 173-entity reference table, and aggregated into four economically meaningful types: US bank, Foreign bank (foreign-headquartered banks operating in the US through a New York or London branch, identified by parent-bank home jurisdiction such as Deutsche Bank AG, BNP Paribas, Sumitomo Mitsui, or UBS), Nonbank (independent nonbank lenders, asset managers, and insurance companies), and Other (BDC investment-adviser affiliates, trust companies, and central-bank discount-window borrowings). SEC-registered broker-dealers are reclassified into the type of their parent: Wells Fargo Securities, Morgan Stanley Senior Funding, Citigroup Global Markets, BofA Securities, Goldman Sachs Lending Partners, Truist Securities, and similar US-bank subsidiaries are folded into US bank; BNP Paribas Prime Brokerage and CIBC World Markets are folded into Foreign bank; KKR Capital Markets and other independent broker-dealers are folded into Nonbank. Panel A reports the count of distinct lender entities by type, total lender appearances (one observation per BDC-facility-quarter-lender combination), and admin-agent-only appearances; Panel B lists the top 15 administrative agents ranked by facility-quarters served as admin agent, with their type, the number of distinct BDC borrowers (# BDC), the number of distinct credit-line agreements (# CLID), and total quarterly observations.

Panel A: Composition by lender type

Lender type	# Lenders	Total Obs	Admin Obs
US bank	97	7,243	4,811
Foreign bank	26	2,363	2,070
Nonbank	33	1,136	867
Other	15	264	220
Total	171	11,006	7,968

Panel B: Top 15 administrative agents (by facility-quarters served as admin agent)

Lender (canonical name)	Type	# BDC	# CLID	Total Obs
JPMorgan Chase Bank, N.A.	US bank	47	74	895
Deutsche Bank AG, New York Branch	Foreign bank	29	47	561
ING Capital LLC	Nonbank	23	26	518
Citibank, N.A.	US bank	26	36	480
Sumitomo Mitsui Banking Corporation	Foreign bank	31	44	436
Wells Fargo Bank, N.A.	US bank	30	36	418
Wells Fargo Securities, LLC	US bank	13	21	300
SunTrust Bank	US bank	9	12	265
BNP Paribas	Foreign bank	19	26	256
Bank of America, N.A.	US bank	21	31	247
KeyBank, N.A.	US bank	10	17	236
Natixis, New York Branch	Foreign bank	15	21	222
Goldman Sachs Bank USA	US bank	16	17	218
Truist Bank	US bank	15	18	200
Capital One, N.A.	US bank	12	14	176

Table IA.6. Events of BDCs' Bank Credit Lines

This table summarizes contractual events on BDCs' bank credit lines between 2005Q1 and 2025Q4, identified from 10-K and 10-Q narratives. Each event records a discrete change to one credit-line agreement: inception of a new facility, amendment of an existing facility, exercise of an accordion option, termination, or replacement; multi-field amendments are split into one row per field changed. The sample contains 6,584 events across 196 BDC CIKs and 680 distinct credit-line IDs (CLIDs). Panel A reports the composition of all events by event type; Panel B reports amendment events by the contractual term modified, alongside the share of those amendments that move the term up versus down (parsed from the recorded old- and new-value strings for numeric terms, with maturity and revolving-period columns denoting extension versus shortening), and direction is not applicable for categorical fields such as benchmark, lender syndicate, subsidiary borrower, covenants, and facility name. Time-series patterns in event composition are reported in Figure IA.7.

Panel A: Events by type

Category	Count	Share (%)
Inception (new credit line)	2,224	33.8
Amendment (term change)	3,729	56.6
Accordion exercise (commitment increase via accordion)	327	5.0
Termination (facility ended)	205	3.1
Replacement (facility refinanced into new agreement)	99	1.5
Total events	6,584	100.0

Panel B: Amendments by field changed (with directional split)

Field changed	Count	Share (%)	% increase	% decrease
Credit-limit (commitment size)	1,294	34.7	71.3	28.1
Spread over benchmark	707	19.0	30.6	61.3
Maturity date	795	21.3	99.1	0.9
Revolving (reinvestment) period	335	9.0	99.7	0.3
Accordion cap (max upside)	164	4.4	92.1	7.1
Benchmark rate (e.g., LIBOR-to-SOFR)	157	4.2	—	—
Lender syndicate composition	101	2.7	—	—
Subsidiary / SPV borrower	27	0.7	—	—
Covenants	134	3.6	—	—
Facility name / parties	15	0.4	—	—
Total amendment-field changes	3,729	100.0		

Table IA.7. Accordion Feature in BDCs' Bank Credit Lines

This table summarizes the prevalence and use of accordion features in BDCs' bank credit lines between 2005Q1 and 2025Q4. An accordion feature gives the borrower the right to upsize the facility (above the base credit limit) without negotiating a new agreement, subject to a contractually specified accordion cap. A credit line is classified as having an accordion feature if its disclosure history contains an explicit accordion-exercise event, an amendment to the accordion cap, or a mention of an accordion option in the inception text. The sample is 680 distinct credit lines (CLID identifiers) across 198 BDCs. Panel A reports prevalence and exercise activity at the CLID level; Panel B reports the distribution of the accordion-cap to credit-limit multiplier across the 137 accordion-having CLIDs for which both values are extracted with consistent units (latest values per CLID, normalized to \$millions, with plausibility filter [0.5, 20] applied to drop unit-mismatch outliers).

Panel A: Prevalence and exercise activity

	Count	Share (%)
Total credit lines (CLIDs)	680	100.0
CLIDs with accordion feature ever	206	30.3
of which: exercised accordion at least once	108	52.4*
Total accordion exercises in sample	327	
mean / median / max exercises per active accordion	3.03 / 2 / 16	

Panel B: Accordion cap / credit limit multiplier

	Mean	SD	Min	P10	P25	P50	P75	Max
Multiplier ($N = 137$ CLIDs)	1.42	0.70	0.68	1.00	1.00	1.14	1.56	4.80